LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy

A Capstone Project presented to the
College of Computer Studies of
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

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ABSTRACT

Title : LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

The growth of digital learning has made independent study an important part of modern education. Students increasingly use review tools, flashcards, and quiz platforms to support lessons outside the classroom. Many of these platforms, however, still follow a one-size-fits-all approach, giving every learner the same activities and materials regardless of individual progress. Recent literature identifies personalized learning as a shift away from this approach toward adaptive, learner-centered instruction (Fariani et al., 2023; Wang et al., 2025). As a result, students may spend time reviewing concepts they already understand while receiving limited support on topics they find difficult, and activities that do not match a learner's level may reduce engagement and contribute to boredom (Krannich et al., 2025).

Cognitive Load Theory explains why this matters. It suggests that learning is more effective when materials are designed to fit the learner's limited working-memory capacity. In digital environments, excessive text, poorly integrated multimedia, and distracting interface elements can impose irrelevant cognitive load, which reduces learning efficiency and weakens knowledge achievement (Krieglstein et al., 2023; Surbakti et al., 2024). Because learners differ in prior knowledge, attention, and language proficiency, adaptive systems

that adjust content, pacing, and presentation are more effective than fixed ones (Skulmowski & Xu, 2022; Surbakti et al., 2024).

Digital learning technologies have widened access to independent study by letting students learn at their own pace and on their own schedule, and they provide tools such as quizzes, interactive resources, and feedback systems that sustain participation (Alshammary & Alhalafawy, 2023; Josué et al., 2023; Sisouvong & Pasanchay, 2024). Many applications, however, support only one type of activity, so students end up using several unconnected tools. Research has shown that this fragmented use can increase unrelated cognitive load and tool fatigue, which points to the value of integrated environments (Qiu & Qiu, 2026). Combining learning activities and analytics in one platform connects participation, progress monitoring, and feedback, and studies report gains in achievement, self-regulation, and engagement compared with conventional use (Du et al., 2025; Elmabaredy & Gencel, 2024; Yang et al., 2024). Learning analytics strengthens this effect by helping students see their progress, identify weaknesses, and adjust their strategies, particularly when analytics is paired with guidance rather than used as stand-alone tracking (Kleimola et al., 2024).

Artificial Intelligence extends these capabilities. Unlike applications built on fixed activities, AI-powered systems can analyze a student's progress, identify strengths and weaknesses, and deliver personalized learning paths, content, and feedback (Barrera Castro et al., 2024). Research reports that AI-driven adaptive technologies improve learning experiences by adjusting instructional support to

each learner's performance, allowing students to work at their own pace while still receiving guidance (Bayly-Castaneda et al., 2024).

Pasig Christian Academy (PCA) is a private institution established in 1986 offering Preschool, Elementary, and Junior High School education in Pasig City. This study focuses on the Junior High School department, where students develop the independent learning habits they will need in Senior High School. Based on the researchers' requirements analysis and consultation with the school administration, independent learning at PCA relies primarily on official textbooks and teacher-provided materials. These resources support the curriculum effectively, but they are static: every student receives the same content regardless of knowledge level or pace. The needs assessment further revealed that students have difficulty identifying which topics require review, want more engaging study activities, value continuous progress monitoring, and prefer a single platform that brings these functions together.

Rather than replacing existing instructional practice, these findings point to an opportunity to strengthen it. LearnSync AI is proposed as an independent learning platform that complements the school's existing resources through personalized support, adaptive activities, AI-assisted study strategies, gamified review, and learning analytics. The platform integrates Bayesian Knowledge Tracing (BKT) to estimate each student's knowledge and Generative Artificial Intelligence (Generative AI) to produce learning support drawn exclusively from educator-uploaded materials. To preserve curriculum alignment and academic

accuracy, only authenticated educators may upload resources and assign the grade levels that can access them.

Within a single platform, LearnSync AI combines Adaptive Flashcards, Adaptive Test Mode, AI-assisted study strategies such as the Feynman Technique and Free Recall, an AI-powered learning assistant, AI-generated educational games, and learning analytics. BKT continuously estimates knowledge from assessment performance, allowing the flashcard and test modules to adapt to each learner's estimated understanding and the games to scale their difficulty accordingly. Students also see their knowledge progression, study streaks, and earned badges through a personal dashboard. By combining educator-approved resources with BKT and Generative AI, the platform aims to make independent learning more personalized, engaging, and organized while preserving the school's curriculum and instructional practices.

REVIEW OF RELATED LITERATURE

This chapter presents a comprehensive synthesis of local and international literature, research studies, and technological frameworks relevant to the development of LearnSync AI.

Modern Study Challenges and Diverse Student Learners

Developing an effective digital learning platform requires recognizing that students differ in how they learn, process information, and respond to instruction. Learners vary in their prior knowledge, academic abilities, motivation, learning pace, and study strategies, so a one-size-fits-all approach does not effectively support all learners (Barrera Castro et al., 2024; Gligorea et al., 2023). Research on personalized education suggests that learning environments become more effective when they consider individual learner characteristics and provide appropriate support (Barrera Castro et al., 2024). Likewise, adaptive learning has been identified as an alternative to traditional one-size-fits-all learning approaches because it provides learning experiences that respond to individual learner needs and progress (du Plooy et al., 2024; Gligorea et al., 2023).

In addition, studies on adaptive learning emphasize that educational systems that respond to learners' needs and progress can create more engaging and effective learning experiences (du Plooy et al., 2024; Gligorea et al., 2023). As digital learning continues to expand, educational platforms are increasingly

expected to provide flexible and personalized learning experiences that address the diverse needs of students (Barrera Castro et al., 2024).

Fast and Slow Learners

Students differ in the speed at which they process information, understand new concepts, and complete learning activities. These differences influence how learners respond to classroom instruction and digital learning environments. In mixed-ability classrooms, some students quickly master lessons and may become less engaged when instruction progresses too slowly, while others require additional time, practice, and guidance before fully understanding the material. Because learners have different academic needs, using the same teaching pace for everyone may reduce engagement and limit learning opportunities for both high-performing and struggling students. Research on differentiated instruction suggests that learning environments should provide flexible support and varied learning opportunities to accommodate differences in students' abilities and learning pace (Ziernwald et al., 2022).

Likewise, research on educational differentiation indicates that adapting learning experiences to individual learner needs may improve participation, motivation, and academic achievement in diverse classrooms (Eikeland & Ohna, 2022). These findings highlight the importance of designing learning platforms that can support students at different levels of understanding rather than requiring all learners to progress at the same pace.

Digital Overload

The increasing use of digital technologies in education has created learning environments in which students often depend on multiple digital tools and platforms to access learning materials, complete assessments, submit assignments, and communicate with instructors and peers. Although these technologies can improve access and flexibility, managing different interfaces and constantly navigating between separate systems can increase the mental effort required to complete learning tasks. This added effort can reduce the amount of cognitive capacity available for understanding, processing, and retaining new information, especially when students must divide their attention between learning content and platform navigation (Ibrahim et al., 2025; Qiu & Qiu, 2026).

Research on digital learning further suggests that unnecessary complexity in online environments contributes to extraneous cognitive load, meaning mental effort caused by the design of the learning system rather than by the learning material itself, which can reduce learning efficiency and make it more difficult for students to focus on instructional content (Nordin et al., 2026).

Furthermore, studies in learning analytics emphasize the importance of designing organized and accessible digital learning environments that support students through efficient access to educational resources and activities while minimizing unnecessary barriers to learning (Qiu & Qiu, 2026). Taken together, these findings highlight the value of developing integrated learning platforms that

centralize academic resources and learning activities, thereby creating a more organized, efficient, and student-centered learning experience.

Junior High School Students

Digital learning in the Philippines has increasingly required students to study with greater independence, especially in online and modular environments where teacher supervision is reduced and learners must manage their own time, tasks, and study strategies. Research shows that self-regulated learning is important in these settings because students differ in their ability to set goals, structure their learning environment, seek help, and monitor their progress, and weaker self-regulation is associated with poorer learning outcomes (Briones et al., 2023; Calamlam, 2023; Funa et al., 2023). At the same time, many Filipino learners continue to experience barriers such as limited internet connectivity, inadequate resources, difficulty understanding modules, and fragmented learning conditions, which can make independent learning more challenging (Abejuela et al., 2023; Agayon et al., 2022; Bustillo & Aguilos, 2022).

These conditions support the need for learning platforms that provide organized access to resources, adaptive support, and timely feedback. Studies on educational technology further suggest that students' learning experiences improve when digital systems are designed to be accessible, interactive, well-facilitated, and responsive to learner needs, including the use of adaptive learning systems and other technology-based interventions that support

engagement and performance (Batucan et al., 2022; Caliston, 2025; Llacuna et al., 2026).

Thus, developing an integrated and adaptive learning platform for Junior High School students is well grounded in the literature, as this population is still strengthening independent learning habits and can benefit from personalized support, efficient resource access, and technology that aligns with their developmental and educational needs (Barrera Castro et al., 2024; Salva et al., 2022; Villarama et al., 2022).

Students differ in ability, pace, and study habits, so one set of materials for everyone does not serve every learner. The reviewed studies show that adaptive learning works better because it matches each student's level. They also show that students juggle several separate tools, which scatters their materials and adds to their mental effort. Junior High School students face this while still learning how to study on their own, so they benefit most from guidance, adaptive activities, and quick feedback. LearnSync AI answers both problems by combining Generative Artificial Intelligence and Bayesian Knowledge Tracing in one platform.

Analysis of Independent Study Tools and Research Gaps

Independent study tools have become an important part of modern education because they give students flexible ways to review lessons and learn outside the classroom. However, many of these tools support only specific

learning activities, which can lead students to rely on multiple applications and create a fragmented digital learning experience. Evidence shows that multi-platform learning environments increase extraneous cognitive load and tool fatigue, reducing students' ability to focus on actual learning tasks rather than on managing different technologies (Ibrahim et al., 2025; Qiu & Qiu, 2026).

Research also shows that structured instructional design and interactive learning environments can reduce cognitive load, suggesting that more integrated digital platforms would better support independent learning than disconnected single-purpose tools (Nordin et al., 2026). These findings reveal an important research gap in existing independent study tools, particularly the limited evidence on platforms that combine multiple evidence-based learning functions with adaptive support in a single, cognitively manageable environment.

Fragmentation of Current Educational Applications

Educational applications have become widely used to support independent learning by offering tools such as flashcards, quizzes, note-taking, and progress tracking. However, many of these applications are designed to support only specific learning tasks, so students often need to move across several platforms to complete different parts of their study routine. This creates a fragmented digital learning experience in which attention is divided between learning content and managing multiple technologies. Recent research shows that multi-platform learning environments increase extraneous cognitive load,

contribute to tool fatigue, and make it harder for students to focus on meaningful learning tasks rather than navigation, logins, and platform switching (Ibrahim et al., 2025; Qiu & Qiu, 2026).

Research also suggests that fragmented learning resources often lack coherent organization, which further reduces learning efficiency and makes knowledge construction more difficult (Hu & Liu, 2025). Supporting this, integrated resource designs have been associated with better perceived coherence and lower cognitive load, indicating that more unified educational systems may better support independent learning than disconnected single-purpose applications (Lou, 2025).

Limitations of Static Review Systems

Many educational applications still rely on static review systems that give all learners the same materials and review activities, even though learners differ in pace, prior knowledge, and current understanding (Bakri, 2025). This one-size-fits-all structure can be inefficient because conventional review often distributes the same materials uniformly rather than targeting each learner's actual weaknesses (Okubo et al., 2023) Research on personalized learning instead describes adaptive systems as ones that adjust to changing learner needs and recommend resources based on learner characteristics, knowledge level, and preferences (Abu-Rasheed et al., 2023; du Plooy et al., 2024).

Evidence also indicates that adaptive review can improve learning: a personalized review dashboard that recommended insufficiently understood pages produced a significantly higher improvement in quiz scores, while a broader review found that adaptive systems often improve academic performance and engagement compared with non-adaptive environments (Bakri, 2025; du Plooy et al., 2024; Okubo et al., 2023).

Most existing study applications either do one thing only or give every student the same material regardless of pace and prior knowledge. Students are left switching between tools, which adds to their mental effort and makes it harder to focus on the lesson. The literature therefore points to a need for one platform that combines personalized activities, organized materials, and continuous support. LearnSync AI is proposed to fill that gap, bringing Generative Artificial Intelligence, Bayesian Knowledge Tracing, adaptive activities, AI-assisted support, gamification, and learning analytics into a single environment.

Learning Analytics and Student Motivation

Learning analytics has become an important educational tool because learner-facing dashboards give students direct access to information about their progress, helping them monitor performance and identify what they still need to study. Unlike static review systems that provide the same materials to all learners, dashboards can support self-regulated learning by making progress

visible and guiding reflection, feedback, and next actions (Paulsen & Lindsay, 2024; Yang et al., 2024). Experimental and quasi-experimental evidence suggests that these systems can improve students' awareness of their learning, self-efficacy, engagement, and academic outcomes when the dashboards are intentionally designed to support motivation and regulation rather than simply display data (Fleur et al., 2023; Yang et al., 2024).

Research also indicates that dashboards work best when grounded in motivational theory and designed to help learners make autonomous, strategic decisions, highlighting the limitations of static one-size-fits-all review approaches that do not adapt to students' ongoing progress (Lee & Kim, 2024).

Learning Progress

Visualizing learning progress helps students engage in self-monitoring because it makes their current standing and learning trajectory visible during the learning process, not only after formal grading events (Paulsen & Lindsay, 2024; Ponomariovienė et al., 2025). In competency-based settings, progress monitoring is tied to feedback, reflection, and self-regulation, with regular feedback helping students understand achievement and adjust strategies to reach learning goals (Ponomariovienė et al., 2025).

Learner-facing dashboards have increasingly shifted from teacher-oriented analytics tools to student-focused systems intended to support learning directly (Paulsen & Lindsay, 2024). A systematic review found that

monitoring, awareness, and comparison are the most common dashboard affordances, but more recent designs increasingly include feedback, reflection, and goal-setting features that better support learning action (Paulsen & Lindsay, 2024).

Evidence from higher education also suggests that dashboards can increase engagement with self-assessment systems. In a study of 194 students, those with dashboard access interacted more with a computerized classification testing system and reported higher confidence than students using the same system without a dashboard (Ifenthaler & Şahin, 2023). The dashboard in that study displayed knowledge-related indicators such as the number of mastered subject areas, correct and incorrect responses, and total attempts, showing how progress visualizations can make formative assessment more usable for students across a semester (Ifenthaler & Şahin, 2023).

Study Consistency and Behavioral Engagement

Study streaks and digital badges are features that can encourage students to build consistent learning habits by recognizing regular participation and achievement. A study streak tracks consecutive days of activity, while digital badges mark milestones such as completing lessons or meeting learning goals. These kinds of gamified feedback and recognition features tend to strengthen motivation, participation, and sustained involvement in digital learning environments (Fauzi et al., 2025; Jun & Lucas, 2024; Sappaile et al., 2025).

Recent evidence also suggests that self-monitoring and feedback-linked gamification can support self-regulated learning and persistence. In one study, learners in a badge-sharing condition showed higher self-regulation and a higher completion rate than controls, with persistence of 70% versus 56% (Xu & Richardson, 2024). Broader reviews further indicate that badges, points, and feedback can promote active participation, progress tracking, and long-term involvement when they are meaningfully aligned with learning goals (Jun & Lucas, 2024).

Overall, current research supports the view that study streaks and digital badges can help promote regular participation, stronger engagement, and more consistent independent learning habits, although their effects depend on careful design and can be mixed for some learners (Adanyin, 2024; Balci et al., 2022; Thakarkar et al., 2025).

Time-on-Task and Self-Regulated Learning

Time-on-task information in digital learning environments can support students' self-regulated learning by making study behavior visible and easier to evaluate during learning rather than only afterward. Clickstream evidence shows that online learners differ in time investment and time-use patterns across performance groups, suggesting that study time is a meaningful indicator of how students manage learning activities (Cao et al., 2022). More broadly, research on self-regulated learning tools shows that monitoring, goal setting, and

self-evaluation are among the most commonly supported processes in online environments, which fits the use of time-on-task displays as a way to help learners track and adjust their behavior (Perez-Alvarez et al., 2022).

Recent dashboard studies further suggest that students value analytics displays for observing study progress and appraising their learning behavior, although their impact depends on clear reference points and integration into everyday learning processes (de Vreugd et al., 2024; Kleimola et al., 2024). Overall, displaying time-on-task appears to promote learner awareness, responsibility, and more intentional study habits, but the broader intervention literature shows that effects are mixed when analytics tools are not linked to planning and reflection (Heikkinen et al., 2022).

Across these studies, letting students see their own learning data supports motivation and self-managed study by making progress and habits visible. Progress displays, study-time tracking, and rewards such as streaks and badges help learners keep steady routines and spot their own weak points, which matters most when no teacher is watching. LearnSync AI therefore gives each student a dashboard showing real-time knowledge progress, study streaks, and digital badges, turning their own activity into feedback they can act on.

Using Artificial Intelligence for Personalized Student Support

Artificial intelligence improves personalized student support by adapting instructional content, pacing, and feedback to individual learner needs rather

than delivering the same materials to all students. AI-based educational systems use learner data to build profiles, identify knowledge gaps, and recommend suitable learning activities, which helps students progress at their own pace and receive more targeted support (Barrera Castro et al., 2024; Murtaza et al., 2022).

Recent studies also suggest that adaptive feedback and intelligent tutoring systems can improve engagement, conceptual knowledge, and satisfaction compared with static or traditional approaches, although effectiveness depends on design quality and subject context (Naseer & Khawaja, 2025; Sajja et al., 2023; Villegas-Ch. et al., 2025). At the same time, the literature consistently notes concerns about privacy, algorithmic bias, infrastructure, and the need to integrate AI with human teaching rather than replace it (Bahroun et al., 2023; Lin et al., 2023; Yarlagadda, 2025).

AI in Modern Classrooms

AI in modern classrooms has helped shift instruction from a uniform model toward adaptive learning environments that respond to individual learner needs. Recent reviews describe this transition as a departure from traditional, one-size-fits-all teaching, with AI enabling systems to analyze student performance, preferences, and interaction data in order to tailor content, pacing, and instructional support (Lin et al., 2023; Rane, 2024; Yarlagadda, 2025). Adaptive platforms and intelligent tutoring systems use machine learning, natural language processing, and learning analytics to provide real-time feedback,

targeted recommendations, and personalized learning pathways, which can make instruction more flexible and student-centered (Akintola et al., 2025; Gligorea et al., 2023; Murtaza et al., 2022).

Empirical studies also report gains in engagement and academic outcomes when AI-driven systems are implemented well, including a 25% improvement in grades, test scores, and engagement in one higher education experiment and higher post-assessment scores across mixed educational settings (Naseer et al., 2024; Sari et al., 2024; Villegas-Ch. et al., 2025). At the same time, the literature is clear that successful classroom integration depends on educator training, infrastructure, ethical safeguards, and maintaining AI as a support for human teaching rather than a replacement for it (Barrera Castro et al., 2024; Halkiopoulos & Gkintoni, 2024; Walter, 2024).

AI in Finding Student Learning Gaps

Artificial Intelligence has become an important tool for identifying students' learning gaps and providing personalized academic support. Unlike traditional assessments that often detect difficulties only after periodic testing, AI-driven adaptive learning systems can continuously analyze student performance and learning behaviors to identify weaknesses in real time and provide immediate support (Akintola et al., 2025; Khanipoor & Karimian, 2024; Onesi-Ozigagun et al., 2024).

These systems can then recommend targeted learning activities, adjust content difficulty, and deliver personalized feedback, helping students focus on areas that need improvement and strengthen understanding during independent learning (Akavova et al., 2023; Gupta, 2024).

Artificial Intelligence supports education by personalizing lessons, adjusting content to how a student is performing, and finding the gaps that need more work, things a single fixed lesson for the whole class cannot do. LearnSync AI uses this to adapt its activities as a student progresses and to point out the concepts that need reinforcement, so independent study stays matched to the learner.

Combining Gamification and Evidence-Based Learning Strategies

Gamification and evidence-based learning strategies appear to support different but complementary aspects of learning in digital platforms. Gamification primarily improves motivation, engagement, and participation, while evidence-based strategies such as retrieval practice, repeated practice, and immediate feedback are more directly tied to stronger retention and conceptual learning (Andretta et al., 2025; Nylén-Eriksen et al., 2025; Trayco et al., 2025). Recent reviews and empirical studies suggest that the most effective gamified environments are those that deliberately align game mechanics with sound instructional design, feedback, and opportunities for active recall rather than

using rewards as isolated features (Dahalan et al., 2023; Lee et al., 2025; Zatserkovnyi, 2025).

Emerging work further suggests that AI-enhanced gamified systems may strengthen this integration by adapting difficulty, feedback, and reward timing to learners' progress while supporting long-term retention and motivation (Aswini et al., 2026; Kassenkhan et al., 2025; Kumar, 2026).

Gamification Procedures

Gamification has become an effective teaching approach that can improve student engagement, motivation, and learning outcomes by incorporating game elements into educational activities. Recent studies show that gamification can increase learners' participation, motivation, and performance in both classroom and digital learning environments (Dahalan et al., 2023; Dehghanzadeh et al., 2023; Hellín et al., 2023; Khaldi et al., 2023; Qudsi, 2024). In digital learning platforms, gamified activities provide students with interactive opportunities to review previously learned concepts while sustaining engagement during independent study.

Research also suggests that gamification is most effective when game elements are aligned with sound instructional design, meaningful feedback, and clear educational objectives rather than functioning only as entertainment or reward systems (Dehghanzadeh et al., 2023; Khaldi et al., 2023; Kode, 2025; Trayco et al., 2025). However, the literature also notes that effects can vary

depending on learner characteristics, novelty effects, and the quality of implementation (Ratinho & Martins, 2023; Ukgoda, 2025).

Shared Mechanism

Crossword puzzles, matching cards, and timed trivia questions are connected because they all use the same basic learning process: students must remember and retrieve information, not just reread it, and gamification research shows that this kind of activity can improve learning, motivation, and participation (Dahalan et al., 2023; Monzon & Hays, 2025). They also share common game features such as challenge, immediate feedback, and clear goals, which help keep students engaged during review activities (Al-Khresheh, 2025). Crossword studies show that puzzles improve terminology recall, vocabulary retention, and active participation because students solve clues by remembering meanings and spellings (Hsu et al., 2023; Jamal et al., 2026; Ramadhani & Shofiyuddin, 2025). Matching card activities show a similar pattern because students must connect related ideas from memory, and card-based learning has been found to improve retention, interest, and confidence (Yu et al., 2025).

Timed trivia is related to the other two because it also depends on fast recall, repeated practice, and immediate correction of answers, which are features linked to better memory and attention in gamified learning (Al-Khresheh, 2025). In simple terms, the three games are different in format, but they do the same job: crossword strengthens word and term recall, matching cards

strengthen associations between related concepts, and timed trivia strengthens quick retrieval and feedback-based review (Hsu et al., 2023; Jamal et al., 2026; Yu et al., 2025).

Crossword

Crossword puzzles have become a useful educational activity for reinforcing knowledge and improving students' understanding of key concepts. Unlike traditional review methods that rely only on reading or memorization, crossword puzzles encourage learners to actively recall information by identifying and connecting terms with their corresponding definitions or clues. This process promotes deeper cognitive engagement while making learning more interactive and enjoyable.

Research suggests that crossword puzzles can improve students' understanding of key concepts and support better performance in objective assessments by encouraging active recall during learning activities (Arnold et al., 2024). These findings suggest that incorporating crossword puzzles into digital learning platforms can strengthen concept retention while increasing student engagement during independent study.

Matching Card Games

Matching card games are widely used as interactive learning activities that promote active participation and reinforce students' understanding of concepts. By requiring learners to match related terms, definitions, images, or concepts, these activities encourage active recall and strengthen associations between pieces of information rather than relying on passive reading.

Recent research found that matching card-based learning materials significantly improved students' vocabulary knowledge and learning outcomes by providing an engaging and interactive learning experience (Febrianti & Shofiyuddin, 2025). Because matching activities encourage learners to repeatedly recall and associate related concepts, their underlying principles can be adapted to reinforce foundational knowledge in digital learning environments.

Timed Trivia Questions

Timed trivia questions are an effective gamified learning activity that combines retrieval practice with interactive quiz-based learning. By requiring learners to answer review questions within a limited time, these activities promote active recall while maintaining attention under structured constraints. Recent evidence shows that game-based quiz platforms improve student engagement, motivation, participation, and academic performance across different educational settings (Dahalan et al., 2023; Nadeem et al., 2023; Özdemir, 2024; Raje & Tamilselvi, 2024).

In addition, game elements such as time limits, scores, leaderboards, and immediate feedback appear to strengthen participation and learning when they are aligned with instructional goals (Jaramillo-Mediavilla et al., 2024; My et al., 2024; Ningsih, 2025; Ratinho & Martins, 2023). Together, these findings support timed trivia questions as an engaging review strategy that encourages active participation while reinforcing previously learned concepts through repeated retrieval and formative feedback (Pellas, 2024; Zainuddin, 2023).

Retrieval Practices

Cognitive retrieval practices are an important component of modern educational psychology because they enhance how students recall and retain learned information. Unlike traditional study methods that rely on rereading or passive memorization, retrieval-based learning requires students to actively recall information from memory, which strengthens long-term retention and improves understanding of concepts (Madan, 2023; Roelle et al., 2022; Serra et al., 2025; Thompson & Hughes, 2023).

Research across educational settings shows that retrieval practice is generally more effective than passive review strategies for improving learning and delayed retention, and it can also support transfer, problem solving, and metacognitive monitoring (Corral & Carpenter, 2025; Muzsnay et al., 2025; Trumble et al., 2023). Retrieval practice can be implemented through strategies such as free recall, low-stakes testing, and structured explanation exercises such

as the Feynman Technique, which encourage learners to reconstruct knowledge actively from memory (Linh & Oanh, 2025; Rajesh & Khan, 2024; Wellmann & Torres Skillicorn, 2023).

Feynman Technique

The Feynman Technique is a learning strategy that helps students develop a deeper understanding of concepts by explaining material in their own words using simple and clear language. Explanation-based learning tasks prompt learners to generate coherent verbal statements that clarify, interpret, and justify ideas, which helps integrate new information with prior knowledge and supports knowledge generalization (Fiorella, 2023). Rather than relying on memorization alone, this kind of learning requires active reconstruction of knowledge, and retrieval-oriented explanation tasks can strengthen both comprehension and long-term retention (Roelle et al., 2023; Roelle et al., 2022).

The Feynman Technique also encourages learners to identify knowledge gaps and revisit source material until they can explain a concept accurately and simply (Linh & Oanh, 2025; Rajesh & Khan, 2024). This process aligns with research showing that generative activities mainly support the construction of coherent mental representations, while retrieval practice mainly supports their consolidation in memory (Obergassel et al., 2025; Roelle et al., 2022; Roelle et al., 2023). Recent experimental evidence suggests that combining generative tasks and retrieval tasks produces better retention than generative tasks alone

and better comprehension than retrieval tasks alone, which fits the logic of the Feynman Technique particularly well (Obergassel et al., 2025; Roelle et al., 2023).

Free Recall

Free recall is a form of retrieval practice that helps students strengthen their memory by encouraging them to remember what they have learned without using notes or other learning materials. Instead of simply rereading information, students actively retrieve knowledge from memory, which strengthens later recall and supports long-term retention (Rea et al., 2022; Serra et al., 2025). Research shows that retrieval practice, including free recall, is generally more effective than passive study methods such as rereading or repeated study for improving delayed learning and memory (Aljabri, 2024; Bego et al., 2024; Klier & Buratto, 2023).

Repeated retrieval can also support recognition, conceptual retention, and transfer when practice is sufficiently repeated and assessed after a delay (Corral & Carpenter, 2025; Huang & Cao, 2026). These findings suggest that free recall is an effective cognitive learning strategy for supporting students' independent learning, although benefits depend on task demands and implementation (Lima & Buratto, 2024; Redifer et al., 2025).

Adaptive Flashcards

Adaptive flashcards are an effective learning tool because they combine active recall with personalized review scheduling. Unlike passive study methods such as rereading notes, flashcards require learners to retrieve information from memory before checking the answer, which strengthens recall and reinforces learned concepts (Huang et al., 2025; Jape et al., 2022). Research on vocabulary learning shows that flashcards can improve recall performance and learning outcomes, and digital flashcards can outperform both paper flashcards and word lists in structured learning contexts (Le & Luong, 2023; Xodabande et al., 2022). Their effectiveness can be further enhanced through spaced repetition, which schedules review sessions over time and increases long-term retention and knowledge transfer compared with non-spaced review or lecture-based study (Price et al., 2024; Santhosh et al., 2024).

Modern adaptive systems extend this approach by adjusting review intervals according to learner performance and item difficulty, which helps focus attention on difficult concepts while reducing unnecessary repetition of well-learned material (Huang, 2025; Jape et al., 2022; Xiao & Wang, 2024).

Adaptive Assessment

Computerized adaptive testing has become an important development in modern educational assessment, shifting traditional fixed testing formats toward more flexible and data-driven evaluation systems. Unlike conventional

assessments where all students answer the same set of questions regardless of ability, adaptive testing systems adjust question difficulty based on learners' responses and performance patterns. Recent research highlights that these systems use real-time performance data and machine learning techniques to select appropriate test items, allowing assessments to better match each student's level of understanding (Zhuang et al., 2024).

In addition, AI-driven quiz generation systems have been shown to support the creation of dynamic assessments that can adjust question difficulty and provide more personalized evaluation experiences (Al-Gburi, 2025). Through these adaptive mechanisms, computerized testing systems aim to improve the efficiency of assessments and provide more accurate representations of student knowledge and learning progress.

Taken together, the reviewed strategies outperform passive review. Gamification keeps students motivated through interactive, goal-driven activities. Active recall and retrieval practice build deeper understanding and longer retention. Spaced repetition and adaptive flashcards concentrate review on concepts a student has not yet mastered instead of repeating what they already know. Adaptive assessment adjusts difficulty to the learner's current level. LearnSync AI combines all four in one platform to support personalized, self-directed study.

Performance Evaluation of Language Models and Inference Systems

The performance of large language models and their inference systems is an important factor in ensuring that AI-powered educational platforms provide fast, reliable, and responsive learning experiences. In digital learning environments, delays in processing user inputs or generating responses can negatively affect student engagement and disrupt learning continuity. As a result, modern LLM systems are designed with optimized inference and serving architectures that improve how models handle multiple user requests efficiently. Research on distributed model serving systems shows that optimized inference techniques can improve system throughput and response efficiency in transformer-based language models (Yu et al., 2022).

In addition, studies on memory-efficient LLM serving highlight that improved resource management and computational optimization help reduce latency and support smoother real-time interactions in large-scale AI applications (Kwon et al., 2023). These advancements are particularly important for educational platforms that rely on generative AI, as they ensure timely feedback, stable performance, and uninterrupted learning support for users.

LLM in Materials and Context

Large language models (LLMs) play an increasingly important role in helping students engage with complex and lengthy learning materials. Early transformer-based language models showed strong capabilities in understanding

and generating human-like text by learning semantic, syntactic, and contextual relationships across language inputs (Shi et al., 2024; Yan et al., 2023). As these models scaled, they developed stronger in-context learning, instruction-following, and reasoning abilities, which expanded their potential use in educational systems as reading, writing, and explanation assistants (Shi et al., 2024; Wang et al., 2024). Although earlier systems were constrained by relatively limited context windows, recent advances in long-context modeling have extended LLM capacity to process much longer inputs, improving their ability to maintain coherence across multi-topic materials, long documents, and extended interactions (Chen et al., 2023; Xu et al., 2023; Yen et al., 2024).

In educational settings, these improvements support more personalized explanations, adaptive pacing, real-time feedback, and better synthesis of dense materials during independent study (Sharma et al., 2025; Wang et al., 2024; Zhui et al., 2024). However, the evidence also shows that effective support depends not only on larger context windows, but on retrieval, context management, and careful instructional design, given persistent risks related to hallucinations, bias, privacy, and student overreliance (Mei et al., 2025; Xu et al., 2023; Yan et al., 2023).

Real Time Feedback

Large language model–based systems play an important role in providing immediate formative feedback within digital learning environments. In traditional

systems, delays in generating responses may interrupt students' focus during active learning tasks such as retrieval practice and review activities. In contrast, modern AI-powered systems are capable of delivering near real-time feedback and personalized guidance, which supports student engagement and sustained participation in learning activities (Kasneci et al., 2023).

Reducing response delays also helps create a smoother interaction between the learner and the system, minimizing disruption during study sessions. This immediate feedback allows students to quickly identify and correct mistakes, which can improve learning efficiency, task engagement, and revision behavior (Meyer et al., 2024).

Model Specialization and Dynamic Routing

The use of multiple large language models within a single system plays an important role in improving the efficiency and responsiveness of AI-powered applications. Instead of relying on a single model to handle all types of tasks, modern systems can distribute requests by assigning them to different models based on task complexity and computational requirements. Research on LLM routing systems shows that queries can be directed to different models depending on expected performance needs, allowing systems to balance quality and efficiency (Ong et al., 2024).

In this approach, simpler or less computationally demanding tasks can be handled by more efficient models, while more complex requests are processed

by higher-capacity models. This routing strategy helps optimize system performance by reducing unnecessary computational load and improving response time, resulting in a smoother and more efficient user experience in real-time applications.

An effective AI-powered platform depends not only on the language model itself but on efficient processing, immediate feedback, an understanding of context, and a sensible division of work among its AI components. LearnSync AI applies these so that it can answer quickly, handle long study materials, return feedback right away, and keep performance steady during use.

Bayesian Knowledge Tracing

Bayesian Knowledge Tracing (BKT) is a probabilistic model used in intelligent tutoring systems to estimate a learner's knowledge of specific skills based on their sequence of correct and incorrect responses. It works by continuously updating the probability that a student has learned a concept, making it useful for modeling knowledge progression in adaptive learning environments.

BKT is commonly used in educational systems due to its simplicity, interpretability, and ability to support dynamic learner modeling without requiring high computational complexity (Bulut et al., 2023).

Tracing Models: BKT, DKT, and PFA

Bayesian Knowledge Tracing (BKT), Performance Factors Analysis (PFA), and Deep Knowledge Tracing (DKT) remain central approaches in educational data mining for modeling student learning progression, but they differ substantially in representation, predictive behavior, and practical use. BKT models student knowledge as binary latent knowledge states updated over time through a Hidden Markov framework, which makes it especially useful for tracking learning sequentially and for producing transparent estimates of knowledge (Bai et al., 2024; Bulut et al., 2023). By contrast, PFA is a logistic alternative to traditional knowledge tracing that predicts future performance from prior successes and failures on one or more knowledge components, making it flexible for items tagged with multiple skills and competitive in some contemporary settings (Lee et al., 2025; Zhang et al., 2025). DKT, in contrast, applies recurrent neural networks to exercise-response sequences and is widely used because it captures complex student interaction dynamics and often achieves strong predictive performance, with later variants such as AT-DKT reporting improved AUC and accuracy across real-world datasets (Lee et al., 2025; Liu et al., 2023).

However, the literature consistently identifies a trade-off between accuracy and interpretability: deep knowledge tracing models can model complex behaviors effectively, particularly when large training datasets are available, but they are often opaque and require substantial tuning, whereas BKT and other transparent models are easier to interpret in educational applications (Bai et al.,

2024; Bulut et al., 2023; Zhou et al., 2024). At the same time, simpler models remain practically important because BKT continues to support personalized intervention through interpretable probabilistic parameters, and recent evidence suggests that PFA can remain competitive while BKT may be more stable than more complex models under concept drift (Bulut et al., 2023; Lee et al., 2025). Overall, recent research suggests that selecting among BKT, PFA, and DKT involves balancing predictive power, transparency, scalability, and robustness, rather than identifying a single best model for all educational contexts (Abdelrahman et al., 2023; Bai et al., 2024; Zhou et al., 2024).

BKT Selection

Among the different knowledge tracing models reviewed, including Performance Factors Analysis (PFA) and Deep Knowledge Tracing (DKT), the researchers selected Bayesian Knowledge Tracing (BKT) as the learner modeling approach for the system. This decision is based on its suitability for adaptive learning environments that require efficient, interpretable, and sequential tracking of student knowledge.

Unlike DKT, which requires large datasets and higher computational resources, and PFA, which focuses on aggregated performance counts rather than step-by-step learning progression, BKT provides a lightweight framework that updates learner knowledge based on individual responses. Its simplicity and transparency make it appropriate for educational applications where clear

interpretation of student progress is important. Therefore, BKT was chosen to support the system's need for continuous, real-time, and interpretable tracking of learner performance in adaptive learning activities such as flashcards and assessments.

Bayesian Knowledge Tracing updates a student's estimated knowledge after every answer, allowing for lucky guesses and careless mistakes so neither moves the estimate too far. It is light enough to run in real time and its estimates can be explained in plain terms. LearnSync AI uses them to set the difficulty of the Adaptive Flashcards and the Adaptive Test Mode, so each student practices at their own level.

SIGNIFICANCE OF THE STUDY

This study proposes LearnSync AI, a personalized learning platform designed to strengthen independent learning through adaptive activities, AI-assisted study strategies, gamified review, and learning analytics. The developed system benefited the following:

Pasig Christian Academy. The system gave the school a centralized, AI-assisted platform that complements its existing independent learning process, integrating adaptive learning, personalized assessment, and verified instructional materials in one secure environment. Because it is designed for asynchronous study at home, it extends the curriculum without disrupting face-to-face operations.

Educators. The system gave educators a secure platform for uploading and managing the official resources LearnSync AI draws from. Because AI-generated support is based only on teacher-approved materials, curriculum alignment and academic accuracy are maintained. The educator dashboard also allows teachers to view student progress and knowledge statistics without significantly increasing their workload.

Junior High School Students. The system gave students a study companion that adjusts activities to their current level. By combining adaptive practice, AI-assisted feedback, gamified review, and personal analytics in one platform, it encouraged engagement in independent learning and helped students monitor their own progress as they prepare for Senior High School.

Researchers. The study developed the researchers' knowledge and skills in software development, educational technology, artificial intelligence, adaptive learning systems, and Bayesian Knowledge Tracing, and gave them practice in designing, developing, implementing, and evaluating an AI-based educational platform.

Future Researchers. The findings may serve as a reference for future work on AI-powered educational platforms, and as a foundation for further study of adaptive learning, Generative AI, Bayesian Knowledge Tracing, and personalized learning technologies in other educational contexts.

STATEMENT OF THE PROBLEM

As Junior High School students progress through increasingly complex academic subjects, independent learning becomes essential in reinforcing classroom instruction and preparing them for the demands of Senior High School. At Pasig Christian Academy, students conduct independent study using educator-provided materials and official textbooks. While these resources support the curriculum effectively, the current process offers limited personalization, adaptive support, and continuous monitoring of progress. To complement it, the researchers developed LearnSync AI, a platform that integrates Generative Artificial Intelligence and Bayesian Knowledge Tracing to provide personalized support, adaptive learning experiences, and continuous progress monitoring. Although the students' independent study covers all of their subject areas, the implementation and evaluation of LearnSync AI in this study used Science as the initial learning content.

This study evaluated the platform's technical accuracy, support for independent learning, the students' learning progress as tracked through Bayesian Knowledge Tracing, and overall acceptability through the following research questions:

1. What are the current independent learning practices and challenges of the Junior High School students of Pasig Christian Academy?

2. What features and functionalities, including the integration of Generative Artificial Intelligence and Bayesian Knowledge Tracing, can be incorporated into the system to address the identified challenges and requirements?

3. How can the correctness of the implemented Bayesian Knowledge Tracing engine be verified against manually computed knowledge estimates?

4. What change was observed in the students' Bayesian Knowledge Tracing (BKT) knowledge estimates recorded by LearnSync AI across their practice opportunities during the implementation period?

5. What is the level of acceptability of LearnSync AI as evaluated by the intended users, specifically Junior High School students, the educator, and information technology practitioners, in terms of the following eight ISO/IEC 25010 software quality characteristics:

    - Functional Suitability
    - Performance Efficiency
    - Compatibility
    - Usability
    - Reliability

- Security
- Maintainability
- Portability

SCOPE AND LIMITATIONS

This study focused on the design, development, implementation, and evaluation of LearnSync AI, an independent learning platform that supports Junior High School students through adaptive learning, Generative Artificial Intelligence (Generative AI), Bayesian Knowledge Tracing (BKT), gamification, and learning analytics in a single platform.

Scope of the Study

The platform complements the school's existing learning process by supporting independent study outside regular classroom instruction. It is not intended to replace classroom teaching or disrupt scheduled activities, and serves instead as an extension of the curriculum using educator-approved materials.

An educator module allows authenticated educators to upload and manage official learning resources and to view student learning statistics. These uploaded materials are the exclusive knowledge base for every AI-assisted function, which keeps generated content aligned with the curriculum.

LearnSync AI integrates the classical four-parameter BKT model to estimate each learner's knowledge from their performance in Adaptive Flashcards and Adaptive Test Mode. These estimates personalize learning activities and adjust the question selection and difficulty of AI-generated games, namely Crossword, Term Matching, and Timed Trivia.

Generative AI further supports independent learning through AI-assisted study strategies such as the Feynman Technique and Free Recall, personalized feedback, and an AI-powered learning assistant that answers questions based only on educator-uploaded resources.

The study evaluates LearnSync AI using the applicable ISO/IEC 25010 Software Quality Model, specifically Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

The implementation and evaluation involved 49 Junior High School students from Grades 7 to 10 of Pasig Christian Academy during the Academic Year 2026–2027, representing 50 percent of the 98 students in the accessible population, together with one (1) educator who managed the learning materials and ten (10) information technology practitioners who evaluated the platform from a technical standpoint. Although the respondents came from four grade levels, they were analyzed as a single study population, and comparative analyses among grade levels are outside the scope of the study. The evaluation

instead described the students' learning progress through the BKT knowledge estimates recorded by the platform across their practice opportunities.

Limitations of the Study:

The study ran for three weeks, of which one week was actual use. Of the 49 selected students, 46 opened the platform and three did not, so those three carry no recorded estimate. Three weeks covers part of a curriculum, not a school year, so the findings describe short-term progress rather than long-term retention or lasting change in academic performance.

The platform supports concept-based learning and review. Specialized Mathematics instruction, meaning symbolic computation, equation solving, graphing, and step-by-step problem solving, falls outside its scope, so where Mathematics was a grade level's most difficult subject, the study used another subject that grade level identified and that suits concept review.

Knowledge estimates come only from Adaptive Flashcards and Adaptive Test Mode, describe performance inside LearnSync AI, and were never compared with school examinations or grades. The games, the Feynman Technique, and Free Recall read the estimate or return feedback without changing it, so a student who used mainly those features produced little knowledge data, and the records track use of the assessment modules rather than of the whole platform. The classical four-parameter BKT model applies a learning transition after every practice opportunity, so estimates tend to rise with

practice. A rising estimate is therefore not by itself evidence of learning, and the study read the variation between trajectories, the differences across topics, and the proportion reaching the threshold in Appendix A instead. Because the model holds no prior evidence about an individual, every student begins at the same default probability, so the estimates become personal only as practice accumulates.

There was no control group, so the recorded progress cannot be credited to LearnSync AI alone; classroom instruction, study outside the platform, prior knowledge, and other resources may also have contributed.

Every AI feature works only from educator-uploaded resources and students cannot upload materials, so the explanations are only as good as what the educator provides. The platform needs a stable internet connection and accepts plain-text PDFs up to 20 MB and image-based or scanned documents up to 20 pages, read through Optical Character Recognition, whose accuracy falls on blurred, low-resolution, poorly lit, or skewed pages and takes the quality of AI-generated support down with it. The platform does not support offline use, handwritten text recognition, voice interaction, payment processing, automated billing, or enterprise-level server scalability.

The platform depends on third-party services: the Gemini API for AI generation and evaluation, and Supabase for storage and semantic retrieval. Preliminary testing traced slow responses to latency in the Gemini API while database and hosting performed normally. Responsiveness therefore rests with

providers outside the researchers' control, and generation can fail when a provider is unavailable or request limits are reached.

CHAPTER II

INTRODUCTION

This chapter presents the methodology used to design, develop, and evaluate LearnSync AI, a web-based platform built to support the independent learning of Junior High School students, with an accompanying Educator Portal that gives teachers control over the materials their students access. The sections that follow describe the system's technical design, development workflow, algorithms, and data structures.

The application was designed around two user roles. The educator uploads and manages learning materials, ensuring all content is reviewed by a qualified teacher before it reaches students, and monitors progress through a class-level analytics dashboard. The student accesses these materials and uses the platform's AI-assisted features, including Adaptive Flashcards, Adaptive Test Mode, the Feynman Technique, Free Recall, AI-generated games, and a personal dashboard. Bayesian Knowledge Tracing (BKT) continuously estimates each student's knowledge and uses the result to personalize activity difficulty. The chapter closes with the implementation and evaluation procedures: a three-week deployment at Pasig Christian Academy, evaluated through an instrument based on the ISO/IEC 25010 software quality model.

RESEARCH DESIGN

This study used a developmental research design with a quantitative approach. In educational technology, developmental research focuses on the systematic creation, testing, and evaluation of instructional tools to ensure they are stable, functional, and effective (Angeles & Cortez, 2023). This design moved the project from software planning to a working application tested by its target users, and allowed the team to translate educational needs into functional features while collecting objective measures of performance.

The process was structured into three phases. In analysis and design, the team identified the study challenges faced by Junior High School students and the supervisory needs of the educator, then used these findings to design the Student and Educator Portal layouts, the study space architecture, the BKT routing rules, and the PlayMe gamification mechanics. In development, the team configured the Supabase relational database with Row-Level Security policies, programmed the generative AI parser, implemented the real-time BKT scoring engine, and built the educator analytics dashboard. In evaluation, the finished platform was deployed over three weeks at Pasig Christian Academy, where the educator uploaded materials and students used the platform independently.

Two complementary methods produced the quantitative data. The first was the extraction of system-generated logs from students' interactions with Adaptive Flashcards and Adaptive Test Mode, where the BKT algorithm computes the probability of learner knowledge from assessment performance.

The second was a software quality evaluation based on ISO/IEC 25010, in which participants completed a four-point Likert questionnaire after using the platform. Responses were analyzed using descriptive statistics to determine the level of acceptability across the eight quality characteristics.

ARCHITECTURAL FRAMEWORK

LearnSync AI follows a dual-portal, three-tier model. The Educator Portal handles material upload, grade-level access control, and knowledge monitoring; the Student Portal handles independent study, adaptive assessment, and gamified review. Information flows down through the Presentation, Application, and Data Tiers when a user performs an action, and back up when the system returns a result.

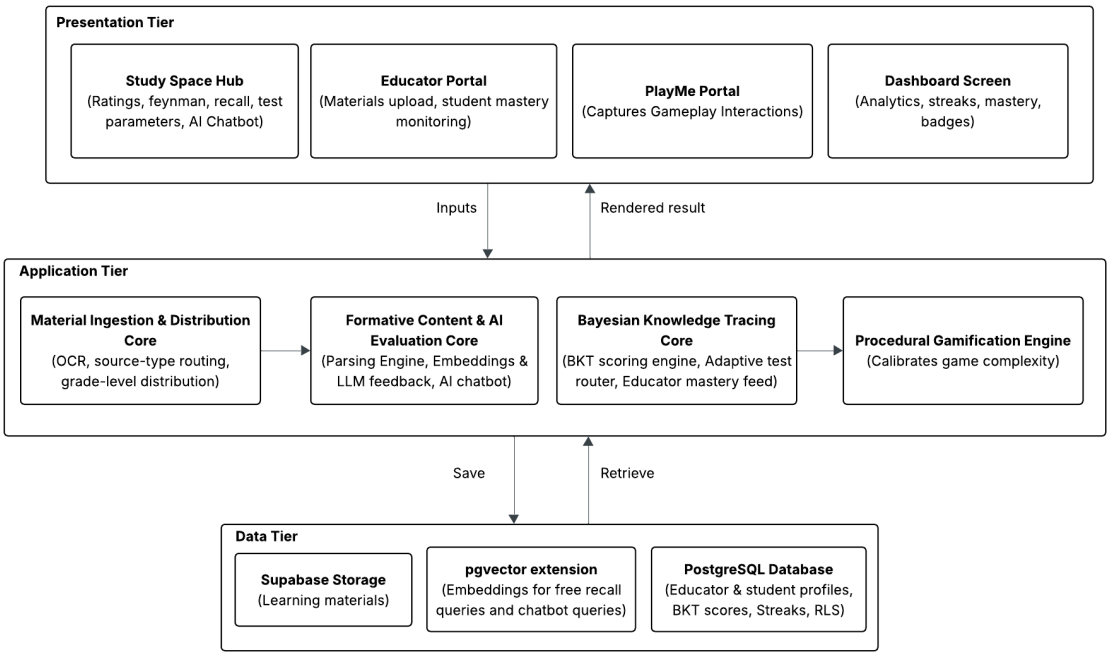
1. **Presentation Tier -** The interface layer, with four parts. The Educator Portal allows upload of plain-text PDF modules or images of printed pages, selection of the grade level that may access them, and a real-time view of student knowledge. The Study Space Hub is where students open materials, rate flashcards, complete Feynman Technique and Free Recall exercises, configure practice tests, and use the AI chatbot. The PlayMe Portal captures game interactions. The Dashboard Screen displays personal progress, streaks, knowledge curves, and badges.

2. **Application Tier -** The processing layer, with four parts. The Material Ingestion and Distribution Core routes uploads by source type: plain-text PDFs pass directly to the parsing engine, while image uploads are first converted through Optical Character Recognition (OCR), which allows lessons that exist only in printed textbooks to enter the same pipeline. The Formative Content and AI Evaluation Core extracts key terms, evaluates Free Recall and Feynman responses using vector embeddings and LLM feedback, and grounds chatbot responses in the uploaded resource. The BKT Core takes flashcard ratings and test answers, calculates per-concept estimates, routes adaptive test difficulty, and feeds aggregated data to the dashboards. The Procedural Gamification Engine reads the knowledge score to calibrate game difficulty.

3. **Data Tier -** The storage layer, running on Supabase. Supabase Storage holds the raw uploaded files. The pgvector extension stores the vector embeddings used for Free Recall similarity checks and chatbot retrieval. The PostgreSQL database holds educator and student profiles, grade-level access rules, BKT knowledge scores, and study streaks, with Row-Level Security (RLS) isolating each student's data and restricting educators to their selected grade level.

Figure 1.

Architectural Framework



Note. The figure shows the dual-portal, three-tier architecture of LearnSync AI and the flow of data between the Presentation, Application, and Data Tiers

ALGORITHM

This section presents the mathematical foundation of LearnSync AI's cognitive modeling core. The platform integrates the classical four-parameter Bayesian Knowledge Tracing (BKT) model, a probabilistic learner model that remains a widely used and interpretable approach for tracing student knowledge and continues to be validated in current research (Bulut et al., 2023). BKT operates as a dynamic hidden Markov model with observable nodes, tracing a

student's latent knowledge state for a given concept from their sequence of responses without heavy server resources (Abdelrahman et al., 2023).

1. **BKT Mathematical Model -** BKT estimates the probability that a student has learned a concept from their chronological sequence of correct and incorrect answers. Four interpretable parameters govern the model (Bulut et al., 2023): the initial knowledge state $P(L_0)$, the prior probability that a student already knows a concept; the transition probability P(T), the learning rate at which a student moves from not knowing to knowing; the guess probability P(G), the chance of answering correctly without knowing; and the slip probability P(S), the chance of answering incorrectly despite knowing.

a. **Initial Knowledge State** $P(L_0)$: The prior probability that a student already knows or has learned a concept before interacting with any formative study modes or assessments on the platform.
b. **Transition Probability** $P(T)$: The learning rate represents the probability that a student shifts from an unlearned (not-known) state to a learned (known) state after a learning opportunity.
c. **Guess Probability** $P(G)$: The probability that the student has not learned the concept but is still able to answer correctly.
d. **Slip Probability** $P(S)$: The probability that a student who has learned the concept still answers incorrectly due to carelessness, cognitive fatigue, or a reading error.

As the student interacts with the application, their performance at any given learning opportunity *t* is captured as an observable binary variable (obs_t), where an answer is marked as correct ($obs_t = 1$) or incorrect ($obs_t = 0$). When a new observation is captured, the system executes a two-phase Bayesian update. First, the algorithm applies the evidence update phase to calculate the conditional probability of knowledge ($P(L_t \mid obs_t)$ based on the observation:

Formula when the students attempt is correct ($obs_t = 1$) :

EQ (1): $$P(L_t \mid obs_t = 1) = \frac{P(L_t)\,(1-P(S))}{P(L_t)\,(1-P(S)) + (1-P(L_t))\,P(G)}$$

Formula when the students attempt is incorrect ($obs_t = 0$) :

EQ (2): $$P(L_t \mid obs_t = 0) = \frac{P(L_t)\,P(S)}{P(L_t)\,P(S) + (1-P(L_t))\,(1-P(G))}$$

Second, the algorithm performs the transition update phase. This step calculates the projected probability of knowledge for the subsequent learning opportunity $P(L_{t+1})$ by factoring in the learning rate $P(T)$ as the student moves to the next card or question, assuming immediate formative feedback is processed:

EQ (3): $$P(L_{t+1}) = P(L_t \mid obs_t) + (1 - P(L_t \mid obs_t))\,P(T)$$

1.1. **Default Parameter Initialization** - LearnSync AI is newly developed and has no student interaction data, so it cannot calculate parameters from responses. Expectation Maximization, the standard method, requires a large dataset of prior responses (Bulut et al., 2023; Šarić-Grgić et al., 2024). The system therefore uses fixed values, which does not prevent it from working since tracking begins at a student's first interaction.

The knowledge tracing literature does not recommend specific numbers for these parameters, since published values are calculated from each study's own data. What it provides instead are limits on acceptable values. These prevent model degeneracy, a condition in which the model behaves illogically, such as lowering a student's knowledge after a correct answer. A recent systematic review documents three sets of limits still in use (Šarić-Grgić et al., 2024). As reported there, Corbett and Anderson proposed $P(G) < 0.30$ and $P(S) < 0.10$; Beck and Chang proposed $P(G) + P(S) < 1$; and Baker, Corbett, and Aleven proposed $P(G) < 0.50$ and $P(S) < 0.50$ (all as cited in Šarić-Grgić et al., 2024). The values used in this study were chosen because they fall inside these accepted limits.

The following default parameter values were adopted for all newly created concept nodes upon document upload:

a. $P(L_0) = 0.10$. The starting value represents a 10 percent probability that a student already knows a concept when tracking begins. This conservative value was chosen because the platform receives no

prior-knowledge data and performs no diagnostic assessment beforehand, so there is no basis for a higher starting point. Starting low requires knowledge to be established through recorded performance. Of the four parameters, $P(L_0)$ has the shortest-lived effect: because each response multiplies the previous estimate, the starting value has little remaining influence after a few responses.

b. $P(T) = 0.20$. The learning rate gives a student a 20 percent chance of moving from not knowing to knowing at each opportunity. The appropriate rate depends on what an opportunity consists of, and published work sets different values for different activity types (Badrinath et al., 2021; Pardos et al., 2013, both as cited in Šarić-Grgić et al., 2024). Conventional values come from intelligent tutoring systems, where one opportunity is a full guided problem with instructional feedback. In LearnSync AI an opportunity is a flashcard review or a single test item completed without feedback, with students carrying the full self-monitoring burden (Luo & Zhou, 2024), so a moderately conservative rate was adopted. This study acknowledged that applying a single $P(T)$ to both activity types is a simplification.

c. $P(G)$ = Context-Sensitive (Dynamic). In standard BKT the guess probability is one fixed value per concept. LearnSync AI instead varies it by assessment mode, question format, and self-reported confidence. This follows established practice: allowing guess and slip to vary between responses produces more accurate student models than fixed per-concept

values, and question difficulty is among the most frequently studied extensions of this kind (Baker, Corbett, and Aleven, 2008, as cited in Šarić-Grgić et al., 2024). Varying $P(G)$ is mathematically valid because the update formulas accept any value between 0 and 1 for each response. In Adaptive Test Mode the values are not estimated but calculated: for each question format, $P(G)$ is the chance of answering correctly by random guessing, determined by the number of answer choices.

In Adaptive Flashcard Mode the value follows the student's confidence rating. Easy assigns $P(G_easy) = 0.05$, reflecting high confidence that the answer came from genuine knowledge. Medium assigns $P(G_medium) = 0.25$, allowing for partial guessing. Hard assigns $P(G_hard) = 0.10$ and is recorded as an incorrect response; because $P(G)$ appears as $(1 - P(G))$ in the denominator of the incorrect-answer formula, a lower value produces a steeper drop, so 0.10 ensures a self-reported failure of recall reduces knowledge appreciably.

In Adaptive Test Mode, the guess probability is calibrated per question type based on the mathematical probability of arriving at a correct response through random guessing alone. Because the baseline chance of a correct random response is determined directly by the number of available options, these values follow from the item format itself. The following $P(G)$ values were adopted for each question format:

1. Multiple Choice (4 options): $P(G_mc) = 0.25$. With four answer options and one correct answer, the baseline probability of guessing correctly is 1/4 = 0.25.
2. True or False: $P(G_tf) = 0.50$. With only two possible responses, the baseline probability of guessing correctly is 1/2 = 0.50. This higher guess rate ensures that correct responses on True or False items contribute a substantially smaller knowledge boost compared to other question types, reflecting the greater role of chance in binary-choice formats.
3. Identification: $P(G_id) = 0.05$. Open-ended identification items require the student to produce the exact term from memory without any provided options. The probability of guessing the correct term without prior knowledge is negligibly low.
4. Fill in the Blank: $P(G_fib) = 0.05$. Similar to identification items, fill-in-the-blank questions require free recall of specific terms or phrases without cues. The absence of selectable options makes random guessing highly unlikely, justifying $P(G) = 0.05$.
5. Matching Type (4 terms, 6 definitions): $P(G_match) = 0.17$. For any individual term, the probability of selecting the correct definition from the pool of six through random guessing is 1/6 ≈ 0.17, applied per matched item.

The rule preventing this is $P(G) + P(S) < 1$, proposed by Beck and Chang (2007, as cited in Šarić-Grgić et al., 2024) and documented as still in use by the current systematic review of the field.

Every combination used here passes. With $P(S) = 0.15$ throughout, Identification, Fill in the Blank, and Easy-rated flashcards use $P(G) = 0.05$, totalling 0.20; Hard-rated flashcards use 0.10, totalling 0.25; Matching Type uses 0.17, totalling 0.32; Multiple Choice and Medium-rated flashcards use 0.25, totalling 0.40; and True or False uses 0.50, totalling 0.65. Every total is below 1.

Seven of the eight formats also pass both the stricter $P(G) < 0.30$ of Corbett and Anderson and the looser $P(G) < 0.50$ of Baker, Corbett, and Aleven (2008; both as cited in Šarić-Grgić et al., 2024). True or False is the only exception, which cannot be avoided with a two-choice format where random guessing succeeds half the time. The practical result is that a correct True or False answer raises knowledge far less than other formats, correctly reflecting how little it proves.

d. $P(S) = 0.15$. A student who knows a concept still answers incorrectly 15 percent of the time through carelessness, misreading, or fatigue. This value passes the degeneracy rule for every format and the P(S) < 0.50 limit of Baker, Corbett, and Aleven. It exceeds only the strictest limit, P(S) < 0.10 of Corbett and Anderson (1995, as cited in Šarić-Grgić et al., 2024), which was set for tutoring systems with real-time supervision and immediate correction. LearnSync AI is self-directed, where students work alone and are more prone to lapses in attention (Luo & Zhou, 2024).

The model contains two kinds of values. $P(L)$, the knowledge estimate, changes after every response. The four parameters never change; they control

how each update works and remain constant while the system runs. The Bayesian mechanism refines the estimate, not the parameters, which are therefore described here as selected configuration values operating within published limits rather than values the system converges toward. Fixed parameters are standard for BKT implementations operating without historical training data, and the literature treats calculating them from data as preferable wherever sufficient data exists (Šarić-Grgić et al., 2024). Recalculating the parameters from collected interaction data after deployment is identified as future work.

1.2. **Step-by-Step Computation of Student Knowledge** - The following trace shows three consecutive multiple-choice attempts using $P(L_0) = 0.10$, $P(T) = 0.20$, $P(S) = 0.15$, $P(G) = 0.25$. Each attempt applies the evidence update, then the transition update.

Attempt 1 - Correct $obs_t = 1$, prior $P(L_0) = 0.10$:

$$P(L_0 \mid obs_t = 1) = \frac{(0.10)(1-0.15)}{(0.10)(1-0.15)+(1-0.10)(0.25)} = 0.27$$

$$P(L_{0+1}) = 0.27 + (1 - 0.27)\,0.20 = 0.42$$

Attempt 2 - Correct $obs_t = 1$, prior $P(L_1) = 0.42$:

$$P(L_1 \mid obs_t = 1) = \frac{(0.42)(1-0.15)}{(0.42)(1-0.15)+(1-0.42)(0.25)} = 0.71$$

$$P(L_{1+1}) = 0.71 + (1 - 0.71)\,0.20 = 0.77$$

Attempt 3 - Incorrect $obs_t = 0$, prior $P(L_2) = 0.77$:

$$P(L_2 \mid obs_t = 0) = \frac{(0.77)(0.15)}{(0.77)(0.15)+(1-0.77)(1-0.25)} = 0.40$$

$$P(L_{2+1}) = 0.40 + (1 - 0.40)\,0.20 = 0.52$$

Two consecutive correct answers raised knowledge from 0.10 to 0.77, while one incorrect answer reduced it to 0.52, a notable but controlled drop, showing how the slip parameter cushions careless mistakes without erasing accumulated progress.

2. Algorithmic Implementation in Flashcard Mode - Within the study space, spaced repetition manages how the flashcards behave while BKT calculates the student's knowledge of a concept. Flashcards are generated from the educator's uploaded materials; students cannot write cards by hand. A student creates, renames, and deletes decks, but every card is generated against a selected material in one of two formats: the concept's term, which uses no artificial intelligence, or a question written by the large language model. In both, the back of the card is the educator's stored definition reproduced verbatim, so the answer a student is assessed against is always teacher-approved content.

Opening a session loads the knowledge estimates that already exist for that deck's concepts. No record is created in advance; a concept not yet assessed is simply treated as carrying the default prior $P(L_0) = 0.10$ when

read, and a stored record appears only after the first practice opportunity. The student flips a card to reveal its definition and rates it Easy, Medium, or Hard.

Each rating maps to a binary observation, to a guess probability reflecting the student's self-reported confidence, and to a Leitner review interval (Durrani et al., 2024). Easy means the student recalled the card easily: the system records a correct answer ($obs = 1$), applies the low guess probability $P(G_easy) = 0.05$ so the boost reflects genuine knowledge rather than luck, and schedules the card in seven days. Medium means they recalled it but had to think: also a correct answer ($obs = 1$), but the higher $P(G_medium) = 0.25$ yields a smaller, more cautious boost, and the card returns in three days. Hard means they forgot: the system records an incorrect answer ($obs = 0$), applies $P(G_hard) = 0.10$, and returns the card the next day for immediate re-exposure. The progressively longer intervals reserve frequent review for the items the student recalls least well.

A rating updates the knowledge estimate only when the card was due. The system first checks the card's scheduled review date: if it has arrived or passed, the rating is a genuine practice opportunity, so the two-phase Bayesian update is applied, the evidence phase first and then the transition phase, the new $P(L_t + 1)$ is saved, and the card is rescheduled by its interval. If the card is not yet due, the rating counts as extra practice. The student still sees the card and receives feedback, but no estimate and no review date change, and the system reports that the rating was not counted. This preserves the model's assumption

that each observation independently tests latent knowledge; without it, a student could rate a card Hard, re-drill it seconds later having just read the answer, rate it Easy, and drive the estimate upward fastest on the concepts they know least. A newly generated card is due immediately, so the first pass through a deck always counts.

Review dates fall on midnight in the Philippine time zone rather than an exact twenty-four hours after rating, so a student who studies at eight in the evening finds the card waiting after school the next day

3. Algorithmic Implementation in Adaptive Test Mode - Within the adaptive test mode, BKT serves as the main engine for calibrating the difficulty of generated assessment sessions. The student selects one or more question formats (multiple choice, identification, matching type, fill in the blank, or true or false) but not the test's length. The system derives the length from the material: every extracted concept receives exactly one question, distributed across the chosen formats, with matching type allocated only in whole blocks of four terms against six definitions. One question per concept ensures one practice opportunity per concept, since repeated questions on the same concept in one sitting would not be independent observations.

Each response is graded as a binary value (1 correct, 0 incorrect) by the system rather than by the language model, and may be revised for as long as the test remains open. The knowledge estimates update once, at submission. Every

item is then applied in question order as a separate practice opportunity, each using the guess probability of its own format (0.25 for multiple choice, 0.50 for true or false, 0.05 for identification and for fill in the blank, and 0.17 applied per matched item in matching type) together with the fixed slip probability $P(S) = 0.15$, so the knowledge gained from a correct answer scales to how much chance the format allows. An unanswered question is graded incorrect and still applied. The results update the student's concept-specific history in the central database.

When the student requests a new test on the same topic, the system queries that history to scale the difficulty of the new items. The estimates for the material's concepts are aggregated into the document knowledge index M_{doc}, the arithmetic mean of those probabilities, and that index is converted into a difficulty tier that adjusts the generation prompt sent to the language model. At or above 0.85 the model is instructed to generate advanced-tier questions focusing on analytical scenarios, synthesis, and edge cases. Between 0.50 and 0.85 it produces intermediate-tier questions emphasizing conceptual comparisons, multi-concept recall, and practical applications. Below 0.50 it produces simpler, highly supportive retrieval questions focusing on direct definitions and basic conceptual associations.

Both the tier and the knowledge value it was derived from are stored with the generated test, so the calibration of any past session can be verified afterward. This session-based adaptation lets students keep full control over their

test layouts while the system adjusts the cognitive demands of the content to their level.

4. Document knowledge and Gamification Calibration - While BKT tracks individual concepts as isolated, high-precision knowledge components, the system must present a consolidated progress index. When a document is uploaded, the ingestion engine extracts N distinct concept nodes, and the application layer aggregates their independent probabilities using an arithmetic mean. The overall knowledge rate Mdoc is calculated at any timestamp from the most current running BKT metrics:

$$\textbf{EQ (4):}\ M_{doc} = \frac{\Sigma\, P(L_{current,i})}{N}$$

Where M_{doc} is the document knowledge percentage shown as a progress bar on the dashboard; N is the number of distinct concepts extracted from the document; and $P(L_{current,i})$ is the running BKT probability of the i-th concept under the student's profile. Each is computed by applying the evidence and transition equations in chronological order. A concept with no recorded practice opportunity enters the aggregation at the baseline prior $P(L_0) = 0.10$; thereafter each attempt captures the binary response, computes the conditional probability by the corresponding evidence formula, and executes the transition update. The value referenced is the most recently saved $P(L_{t+1})$, so the overview adapts to live progress.

The gamification module uses this score to calibrate the difficulty of generated games (crosswords, term matching, timed trivia), but not as a continuous percentage. On initialization the system converts the current M_{doc} into one of three difficulty tiers, namely foundational below 0.50, intermediate from 0.50 to 0.85, and advanced at or above 0.85. The tier then selects a fixed set of rules for vocabulary complexity and clue difficulty together with a fixed trivia timer of twenty, fifteen, or ten seconds respectively. A student at exactly 50 percent therefore receives intermediate content and a fifteen-second timer. Both the tier and the M_{doc} values behind it are stored with the game for later verification.

The individual concept estimates also determine which concepts a game draws on. Concepts are selected by weighted random sampling in which a lower estimate carries a higher probability of selection, so the student practices what they are weakest at while mastered concepts remain reachable rather than excluded. For this selection alone a concept with no recorded estimate is weighted as though it stood at 0.50 rather than at $P(L_0) = 0.10$, so never-practiced concepts rank below those the student is measurably struggling with and above those already mastered. Because selection is random rather than a fixed ranking, requesting a new game on the same material does not reproduce an identical one. This keeps gamified learning synchronized with cognitive progress while protecting students from frustration.

5. Evaluation in Free Recall - Within the study spaces, the Free Recall tab lets students type everything they remember about a lesson without looking at their notes. To grade this automatically the system compares the meaning of what the student wrote against the lesson's concepts, using Cosine Similarity over high-dimensional vector embeddings, in four steps.

First, the raw text is sent to the large language model, which extracts the discrete ideas the student discussed, producing one item per idea rather than one per paragraph. A point summarizing a whole paragraph would embed as an average of everything in it and would match no single concept closely.

Second, each extracted idea is embedded separately using the same model applied to the educator's material at upload. The lesson's concepts were embedded the same way during ingestion and are stored in the database through the Supabase pgvector extension

Third, each stored concept is compared against every one of the student's idea vectors and the highest comparison is taken as that concept's best match. The comparison measures the angle between two vectors and is computed in the application layer over the vectors retrieved from the database:

$$\textbf{EQ (5):}\ Cosine\ Similarity\ = \frac{A \cdot B}{||A||\ ||B||}$$

where A is a stored concept vector, B a student idea vector, $A \cdot B$ their dot product, and $|A|$ and $|B|$ their magnitudes. A score near 1 means the student's idea points in the same directional meaning as the concept.

Fourth, a concept is marked Covered when its best match reaches the 0.70 threshold and Missed otherwise. The threshold is applied per concept, not to the submission as a whole. The reported score is the coverage ratio: concepts Covered divided by total concepts in the lesson. The Missed concepts are displayed immediately as the specific ideas the student did not mention, with written feedback. A coverage ratio rather than an average of raw similarity scores keeps the figure honest, since even unrelated vectors score about 0.60 in a high-dimensional space and averaging would inflate every result.

The pgvector stores the embeddings and serves them to the application while the similarity itself is computed in the backend, because each submission compares every stored concept against every extracted idea rather than performing one pairwise distance calculation between two fixed vectors.

6. Feedback generation in Feynman Technique - The Feynman Technique helps students learn by explaining hard topics in simple, everyday language, as if teaching a 10-year-old child. The application sends the explanation to a large language model in a single request and receives one structured result covering three checks; the checks below describe what is assessed, not three separate calls.

The Jargon Check scans the explanation against the glossary extracted during document upload. Complex terms used without being broken down, such as “photosynthesis” or “electromagnetism”, are flagged and returned to the

student to explain more simply. The system then removes from that list the name of the concept the student was asked to explain, since naming it cannot itself be jargon.

Analogy Verification checks whether the student used an everyday comparison to make the idea clear, such as comparing an electrical circuit to water flowing through pipes. The system then rejects any comparison drawn from the lesson's own vocabulary, because comparing one idea from the subject to another from the same subject is an in-domain example rather than an everyday analogy and teaches a child nothing.

Finding the Gaps compares the explanation against the concept's stored definition to check whether anything is factually wrong, missing, or incomplete, producing an accuracy score from 0 to 100 that judges only whether what the student asserted is true, regardless of wording.

A second, independent simplicity score from 0 to 100 judges only how well the explanation suits a 10-year-old audience, regardless of correctness; a hollow or circular explanation is penalized here rather than on accuracy. Separating the axes prevents a correct but jargon-heavy explanation from being marked down for accuracy it did not lack, and a simple but mistaken one from being rewarded for clarity alone.

The verdict of Correct, Needs Improvement, or Incorrect is derived by the system from the accuracy score alone: Correct at 80 and above, Needs Improvement from 30 to 79, Incorrect below 30. It is never taken from the model

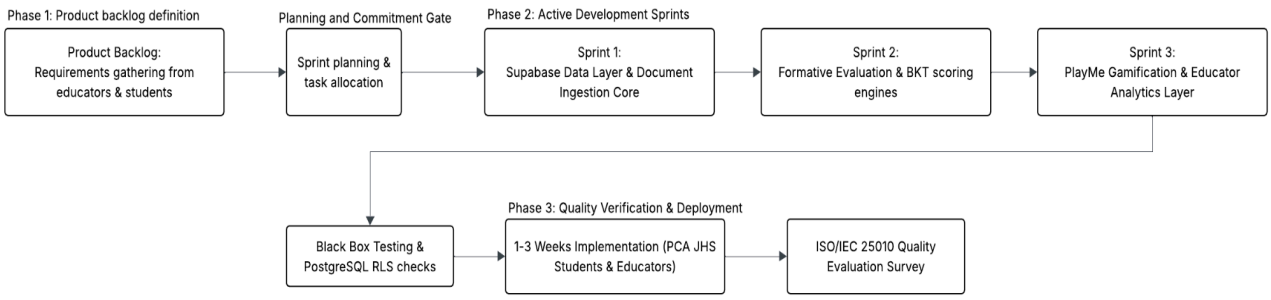
directly, because when asked the model lets the simplicity axis influence its judgment, which would let the badge disagree with the accuracy score beside it. Beyond the scores, the AI writes short, encouraging feedback that points out what the student got right, names factual errors directly, and tells them specifically how to improve

SOFTWARE ENGINEERING PARADIGM

This study used the Agile Scrum framework. Because the platform includes several interconnected features requiring continuous testing and refinement, a flexible methodology was needed. Agile Scrum supports iterative development through short sprints, allowing developers to build, test, and refine while responding to changing requirements (Hodson et al., 2024), and helps identify and resolve issues early, improving software quality and reducing development risk (Strode et al., 2022).

Figure 2.

Agile SDLC



Note. The figure shows the Agile Scrum development cycle followed in building LearnSync AI, from product backlog definition to quality verification and deployment.

The execution of the Agile Scrum framework across the development lifecycle is structured into the following sequential phases:

1. **Product Backlog** - The team established requirements from two role perspectives. The Educator uploads materials, selects which grade level may access them, and monitors knowledge through a class-level dashboard. The Student accesses those materials, engages with four self-paced study options (adaptive flashcards, Free Recall with AI feedback, Feynman Technique explanations, and an AI chatbot), generates adaptive practice tests in five formats that scale with their BKT state, and uses a gamified review engine that adjusts difficulty from the same scores.

2. **Planning and Commitment Stage** - Sprint planning broke the backlog into modular tasks, estimated complexity, set priorities, and established a sprint calendar. This kept cycles focused, prevented scope creep, and prioritized the mathematical models and data security layers before active coding.

3. **Active Development Sprints** - Sprint 1 configured the Supabase PostgreSQL tables for educator and student profiles, grade-level access rules, and materials and embeddings; enforced Row-Level Security; and implemented LLM-based extraction to parse uploads into concepts, definitions, and semantic vectors. Sprint 2 programmed the four-parameter BKT calculations and the adaptive test routing algorithm, established vector-based evaluation for Free Recall, integrated LLM prompt templates for Feynman and chatbot queries, and logged activity timestamps. Sprint 3 coded the backtracking layout algorithm for crossword grids, trivia, and vocabulary matches; built the gamification and streak engine; programmed the educator analytics dashboard; and compiled the backend API gateways with a responsive dual-portal frontend.

4. **Quality Verification & Development** - The application underwent black-box testing and PostgreSQL RLS checks to verify data isolation for both student-to-student and educator access boundaries, then launched for a three-week implementation at Pasig Christian Academy. At the conclusion, an ISO/IEC 25010 quality evaluation survey was administered on a four-point Likert scale.

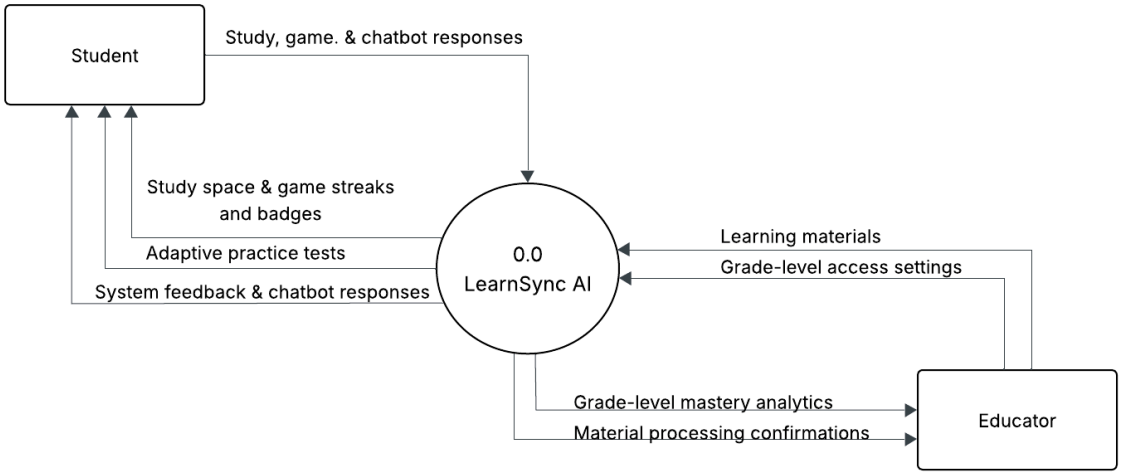
DATA FLOW DIAGRAM

A Data Flow Diagram shows how information moves through the system, from an educator uploading a material to the system returning adaptive tests, feedback, and progress updates to students and knowledge reports to educators. The Level 0 diagram gives a single high-level view; the Level 1 diagram breaks it into the main processes and data stores.

1. **Level 0: Context Diagram -** The context diagram represents the whole application as one process, 0.0 LearnSync AI, with two external entities. The Educator sends learning materials (plain-text PDF modules or images of printed pages) and grade-level access settings, and receives grade-level knowledge analytics and material processing confirmations. The Student sends study, game, and chatbot responses, and receives study space and game streaks and badges, adaptive practice tests scaled to their knowledge level, and system feedback and chatbot responses.

Figure 3.

Level 0 Context Diagram



Note. The figure shows LearnSync AI as a single process together with its two external entities, the Student and the Educator, and the data exchanged between them.

2. **Level 1: Data flow diagram** - The Level 1 diagram contains five processes, three data stores (D1: Study Spaces, Learning Materials and Embeddings; D2: Student Profiles and Knowledge Track; D3: Activity and Streak Logs), and two external entities.

    Process 1.0. Material Upload and Distribution Control, receives materials and access settings from the Educator and stores the raw files and access rules in D1. When a Student opens their study space, it checks D1 for matching access rules and returns the authorized materials.

It also reads aggregated knowledge data from D2 and delivers it to the Educator.

Process 2.0. Document Parsing and Semantic Ingestion, retrieves raw files from D1, extracts keywords, converts the text into vector embeddings, writes the results back to D1, and sends a processing confirmation to the Educator.

Process 3.0. Formative Study and AI Evaluation, handles flashcards, the Feynman Technique, Free Recall, and the chatbot. It receives responses and queries from the Student, reads parsed content from D1, and returns feedback and chatbot responses. AI-generated flashcards are saved to D1. For knowledge tracking it reads the current flashcard knowledge from D2, applies the BKT calculation, and writes the updated score back, logging each session to D3.

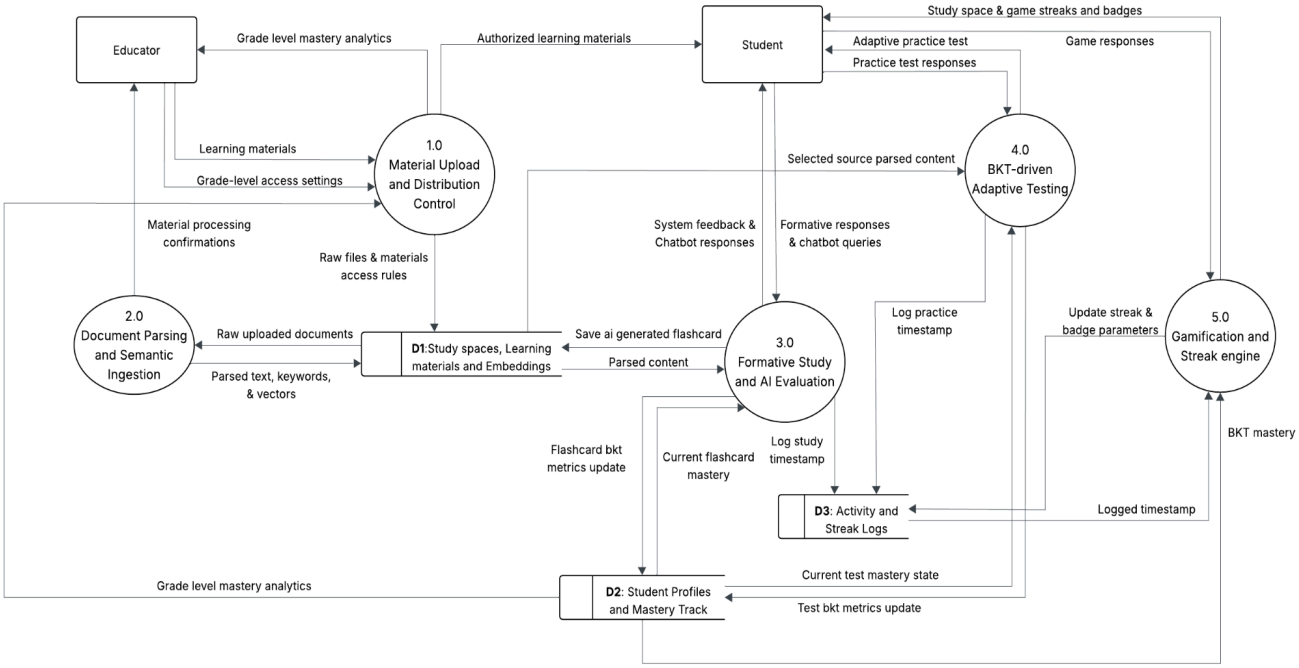
Process 4.0. BKT-driven Adaptive Testing, receives practice test responses from the Student, reads parsed content from D1 and the current test knowledge state from D2, applies BKT, and returns adaptive tests matched to the student's level. Updated scores are written to D2 and session times logged to D3.

Process 5.0. Gamification and Streak Engine, receives game responses, reads the document-level score M_doc from D2 to calibrate difficulty and the logged timestamps from D3 to check daily activity,

updates streak and badge records, and returns streaks and badges to the Student.

Figure 4.

Level 1 Main Process



Note. The figure shows the five processes, three data stores, and two external entities of LearnSync AI, decomposed from the single process in Figure 3.

TOOL TO TEST FOR ACCURACY

To ensure reliability and correctness, the study used a multi-layered validation strategy covering both the algorithmic implementation and the functional behavior of the system. Because LearnSync AI does not use historical interaction data, model training, or data-driven parameter estimation, the

evaluation did not employ machine learning metrics such as Confusion Matrix, Precision, Recall, F1-Score, or Mean Absolute Error. It adopted instead a software engineering verification approach that checks whether the BKT engine and the Generative AI components operate according to their specified formulas and intended behavior.

1. **BKT Algorithm Implementation Verification**

    The BKT module was evaluated through deterministic algorithm verification, confirming that the system produces knowledge estimates mathematically identical to hand-computed values. Because the study uses predefined parameters rather than values learned through Expectation Maximization, the objective was not to assess whether BKT predicts real-world knowledge, but to verify that the code executes the standard four-parameter equations without computational or logical defects.

    The researchers designed structured test cases, each with a known initial state, a specific sequence of binary responses, the default parameters, the context-sensitive P(G) for the interaction mode, and an expected output computed by hand. The cases covered consecutive correct, consecutive incorrect, alternating, and mixed sequences across all three flashcard ratings and all five test question formats. Each case was executed in the live application and compared against the manual

value. Implementation correctness was quantified using the Implementation Correctness Rate:

$$\textbf{EQ (6):}\ ICR = \frac{Number\ of\ Passed\ Test\ Cases}{Total\ Number\ of\ Test\ Cases} X\ 100$$

A case passed if the absolute difference fell within ±0.01, accounting for floating-point variation. An ICR of 100 percent indicates full and correct implementation, and any failure would be documented with its expected value, actual output, and deviation. This verification validates the correctness of the implementation, not the predictive validity of the model itself (Bulut et al., 2023).

2. **Generative AI Functional Verification**

The Generative AI components were evaluated through functional verification rather than model benchmarking, since the study did not fine-tune or retrain the underlying model. The researchers uploaded a controlled learning resource, triggered each AI-powered feature, and evaluated every output against four criteria. Contextual Relevance checked whether the content pertains to the topics and terminology in the uploaded resource. Factual Correctness checked accuracy against the source material. Difficulty Alignment checks whether the content matches the appropriate tier for the student's current knowledge level, confirming that the adaptive pipeline passes the knowledge parameter correctly.

Content Consistency checked whether repeated generations remain stable and non-contradictory; minor phrasing variation was expected, but core facts, terminology, and difficulty were required to hold. Results were reported descriptively.

3. **Black-Box Functional Testing**

Black-box testing validated the behavior of the integrated system against its functional requirements without examining source code or database structure. Educator-side cases covered document upload and content extraction, grade-level access configuration, and the class-level analytics dashboard. Student-side cases covered authentication, access to materials, flashcard generation and spaced-repetition scheduling, adaptive test generation and scoring, game generation and difficulty calibration, BKT tracking and dashboard visualization, Free Recall submission and similarity evaluation, Feynman submission and feedback generation, chatbot interaction, and analytics display. Each case specified a precondition, an input action, and an expected output, and results were recorded in a structured log.

4. **ISO/IEC 25010 Software Quality Model Survey**

To evaluate quality from the end user's perspective, the platform was assessed across the eight characteristics defined by the standard. A structured questionnaire aligned with these characteristics was administered after the implementation period using a Likert scale, and responses were analyzed using mean scores and standard deviations to derive an overall assessment per characteristic. These results complement the technical verification described above.

POPULATION AND SAMPLING

The target population consisted of Junior High School students (Grades 7–10) of Pasig Christian Academy for the Academic Year 2026–2027. As the intended end users of LearnSync AI, these students participated in the implementation and evaluation. Junior High School students were selected because they are at a stage of developing independent learning habits, self-regulated learning skills, and academic readiness, which makes them suitable participants for evaluating the platform and for monitoring learning progress within it.

The study employed proportionate stratified random sampling to select 49 students from the accessible population of 98. Respondents were grouped by grade level, and 50 percent of the students in each level were selected, with the number determined proportionally to the population of that level as presented in

Table 1. Within each grade level the required respondents were drawn through simple random sampling. This technique ensured proportional representation while minimizing selection bias.

Table 1.

Population and Sample Distribution

Grade Level	Population (N)	Sample (n)
Grade 7	16	8
Grade 8	31	15
Grade 9	32	16
Grade 10	19	10
Total	**98**	**49**

Note. N refers to the population of each grade level and n to the sample drawn from it using the proportionate allocation formula in EQ (7).

The sample size for each grade level was determined using the proportionate allocation formula:

$$\textbf{EQ (7):}\ n_h = \frac{N_h}{N}\ x\ n$$

Where:

- n_h = Sample size for each grade level
- N_h = Population of each grade level
- N = Total Population
- n = Total Sample Size

For the educator respondents, one educator responsible for the subject area identified during the requirements analysis as difficult among the participating students was included. For the technical evaluation, ten (10) information technology practitioners were included. All of them hold a bachelor's degree in Information Technology, Computer Science, or a closely related field. They were chosen through purposive sampling, which selects participants for a specific qualification rather than at random, because the study needed evaluators who could judge the platform as software. They were given access to both the Student Portal and the Educator Portal so that they could rate all eight quality characteristics. Unlike the students and the educator, they did not use the platform for classroom learning; they evaluated it as a finished system. Although respondents came from different grade levels, the study did not compare learning outcomes across Grades 7 to 10; all participants were analyzed as a single study population, and the analysis focused on learning progress described through BKT knowledge estimates rather than on differences among grade-level groups.

EVALUATION INSTRUMENT

The evaluation instruments consisted of backend learning analytics and researcher-adapted questionnaires covering learning progress, software quality, and user acceptability.

System-generated knowledge records. The platform automatically logged each student's BKT knowledge estimate per learning topic after every practice

opportunity, together with the number of opportunities completed and the module through which each was generated. These records were produced during ordinary platform use, so no separate assessment was administered. They were extracted at the end of the implementation period and served as the primary instrument for Statement of the Problem No. 4. The correctness of the computed values was established separately through the BKT verification procedure.

Quantitative instrument. Students, the educator, and the information technology practitioners evaluated acceptability using a questionnaire based on the ISO/IEC 25010 Software Quality Model, covering Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Students evaluated student-facing functionalities; the educator evaluated only the educator module, including resource upload and management, grade-level assignment, and student learning statistics; and the practitioners evaluated both portals, rating all eight characteristics from a technical standpoint. The eight-characteristic structure of this questionnaire follows that used in recent Philippine evaluations of educational software (Alan et al., 2026). Respondents rated each statement on a four-point Likert scale that eliminated the neutral option to encourage a definitive evaluation: 4 (Strongly Agree or Excellent), where the functionality consistently performs as intended; 3 (Agree or Good), where it performs satisfactorily with only minor issues; 2 (Disagree or Fair), where it partially meets expectations with noticeable limitations; and 1 (Strongly Disagree or Poor), where it fails to meet expectations.

Qualitative feedback. A final section of open-ended questions allowed students and the educator to give comments, observations, and recommendations, letting them describe their experience and raise concerns that the quantitative items might not capture.

DATA COLLECTION METHOD

Data was collected in four phases: requirement analysis, pre-implementation, implementation, and post-implementation evaluation. Together they carried the needs identified at Pasig Christian Academy through the design, development, implementation, and evaluation of LearnSync AI.

Three groups of people took part, and each answered a different question. In the requirement analysis phase, before anything was designed or built, the researchers interviewed the school principal and the educator and surveyed Junior High School students to establish the current independent learning process, its challenges, and the functional and non-functional requirements of the system. Because that survey existed to find out what students needed rather than to judge a finished system, it was open to every student in Grades 7 to 10 and was not restricted to the evaluation sample. Of the 98 students in the accessible population, 62 completed it, a response rate of 63.27 percent. A separate sample of 49 students, drawn by proportionate stratified random sampling as presented in Table 1, governed the implementation and evaluation phases.

In the pre-implementation phase the researchers obtained permission from the school administration, explained the objectives, procedures, and duration, and showed participants how to access and use the platform.

The implementation phase ran for three weeks. The first covered the presentation of the system to the school principal, the gathering and uploading of learning materials through the Educator Portal, and the creation and distribution of student accounts. The second was the actual usage period, when the 49 selected students were given access to LearnSync AI for independent learning outside regular classroom instruction; 46 of them opened the material and generated records, as reported under Statement of the Problem No. 4, and the other three never did, so they have no recorded estimates. The third covered the wrap-up: the administration of the evaluation questionnaires and the extraction of the system-generated learning records.

The subject was the one the requirements analysis identified as difficult in each participating grade level and suited to the platform's concept-based capabilities. Science was named by 15 of the 62 respondents, or 24.19 percent, as the subject most difficult to understand without a teacher's help, second only to Mathematics and the only other subject named across all four grade levels. The school administration also asked that the uploading and management of learning resources be handled by a single educator so that the class schedules of other subject teachers were not disrupted, so one Science educator was designated for the implementation period. This provided the initial learning

content for evaluation; it did not restrict the platform to one subject area or prevent its future application to others.

Students then used the platform's core features, including Adaptive Flashcards, Adaptive Test Mode, the AI Learning Assistant, the Feynman Technique and Free Recall, AI-generated games, and learning analytics, choosing freely which to use and how often, as self-directed study implies. Throughout the period the system automatically recorded each student's practice opportunities, module-level activity, and BKT estimates per topic, and the researchers monitored the implementation and provided technical assistance when needed.

In the post-implementation phase the evaluators completed the ISO/IEC 25010 questionnaire through Google Forms, along with an open-ended questionnaire gathering comments and recommendations. Fifty-seven (57) evaluators took part: the forty-six (46) students who used the platform, the one (1) educator who managed the learning materials, and ten (10) information technology practitioners who assessed the platform from a technical standpoint. Their responses supported the statistical analysis of software quality and of support for independent learning, and the extracted BKT records were analyzed using descriptive statistics to describe learning progress across practice opportunities.

ETHICAL CONSIDERATIONS

This study was conducted in accordance with established ethical research principles to protect the rights, privacy, and welfare of all participants. Permission was obtained from the administration of Pasig Christian Academy before implementation. Participants were informed of the purpose, procedures, duration, and nature of their participation; participation was voluntary, and participants could decline or discontinue without penalty.

The study complied with the Data Privacy Act of 2012 (Republic Act No. 10173), collecting only the information necessary for implementation and research. Personal information and system-generated learning data, including BKT knowledge records, were treated with confidentiality and used solely for academic research and system evaluation, with no personally identifiable information disclosed in reports, presentations, or publications. All collected information was handled securely and was not shared with unauthorized third parties. To maintain academic integrity, only authenticated educators were authorized to upload learning resources, and access to learning analytics and personal records was restricted to the respective student and authorized educator.

SUMMARY

This chapter presented the methodology used in the development and evaluation of LearnSync AI, covering the research design, system architecture,

algorithm, software engineering methodology, data flow, accuracy testing tools, population and sampling, evaluation instrument, data collection procedures, and ethical considerations.

The study employed a developmental research design using quantitative methods. LearnSync AI was developed using Agile Scrum and integrates the classical four-parameter Bayesian Knowledge Tracing model to estimate students' knowledge. The educator uploads and manages the materials students access; Adaptive Flashcards and Adaptive Test Mode update the knowledge estimates, while AI-assisted strategies, AI-generated games, and personalized activities use those estimates to support independent learning. The Educator Portal provides a grade-level analytics dashboard for monitoring progress in real time.

The study was conducted at Pasig Christian Academy with Junior High School students selected through proportionate stratified random sampling and their assigned educator. The system was implemented over three weeks, and data was collected through questionnaires based on the ISO/IEC 25010 Software Quality Model. The correctness of the BKT implementation was verified against manually computed estimates, and the collected data was analyzed using frequency count, percentage, weighted mean and standard deviation, with Cronbach's alpha for internal consistency and a paired-samples t-test with Cohen's d for the change in knowledge estimates, evaluated at the .05 level, with ethical principles including compliance with the Data Privacy Act of 2012 observed throughout.

Building on the methodology presented in this chapter, the succeeding chapter presents the results obtained from the requirements analysis, development, implementation, and evaluation of LearnSync AI, organized according to the Statements of the Problem, followed by a discussion of these findings in relation to the objectives and research questions of the study.

CHAPTER III

INTRODUCTION

This chapter presents the results obtained from the conduct of the study on LearnSync AI. The objective of the study was to design, develop, implement, and evaluate an independent learning platform that integrates Generative Artificial Intelligence and Bayesian Knowledge Tracing to provide personalized support, adaptive learning experiences, and continuous monitoring of the learning progress of Junior High School students.

The results are organized according to the Statements of the Problem presented in Chapter I. This chapter reports the current independent learning practices and challenges of the students, the features incorporated into the platform to address them, the verification of the Bayesian Knowledge Tracing engine, the change observed in the recorded knowledge estimates, and the level of acceptability of the platform as evaluated by its intended users.

CURRENT INDEPENDENT LEARNING PRACTICES AND CHALLENGES

A researcher-made requirements analysis questionnaire was administered through Google Forms. It consisted of a demographic section, one item identifying the subject perceived as most difficult to understand during unsupervised study, and twelve close-ended statements rated on a four-point Likert scale. Consistent with Chapter II, the scale eliminated the neutral option to

encourage a more definitive response. Table 2 presents the scale and the corresponding mean ranges.

Table 2.

Four-Point Likert Scale and Its Interpretation

Scale	Mean Range	Verbal Interpretation
4	3.26 – 4.00	Strongly Agree
3	2.51 – 3.25	Agree
2	1.76 – 2.50	Disagree
1	1.00 – 1.75	Strongly Disagree

Note. The table shows the four-point scale used in the survey and the score range for each interpretation.

The requirements analysis was conducted during the pre-development phase, before the platform was designed. Because its purpose was to establish requirements rather than evaluate a developed system, the questionnaire was administered across Grades 7 to 10 without restriction to the evaluation sample. Of the 98 students in the accessible population, 62 completed it, a response rate of 63.27 percent. The sample of 49 students drawn through proportionate stratified random sampling, presented in Table 1, governed the implementation and evaluation phases and is reported under Statements of the Problem No. 4 and No. 5.

Responses were analyzed using frequency count, percentage, weighted mean, and standard deviation. Cronbach's alpha for the twelve close-ended

statements was 0.741, above the .70 value conventionally treated as the minimum for internal consistency (Hussey et al., 2025).

Profile of the Requirements Analysis Respondents

Table 3 presents the distribution of the sixty-two (62) student respondents by grade level, together with the corresponding grade-level populations and response rates.

Table 3.

Survey Respondents and Response Rate by Grade Level

Grade Level	Population (N)	Respondents (f)	Percentage of Respondents (%)	Response Rate (%)
Grade 7	16	9	14.52	56.25
Grade 8	31	24	38.71	77.42
Grade 9	32	17	27.42	53.13
Grade 10	19	12	19.35	63.16
Total	**98**	**62**	**100.00**	**63.27**

Note. n = 62. The table shows how many students from each grade level answered the survey. Percentage of Respondents is the share of the 62 respondents who came from that grade level, and Response Rate is the share of that grade level who responded.

Grade 8 accounted for the largest share of respondents at 38.71 percent and also recorded the highest participation at 77.42 percent of its grade-level population, followed by Grade 10 at 63.16 percent, Grade 7 at 56.25 percent, and Grade 9 at 53.13 percent. All four grade levels were represented. Thirty-six

(36) of the 98 students did not complete the questionnaire, so the findings describe the students who chose to respond rather than the accessible population as a whole.

Subjects Identified as Most Difficult During Independent Study

Respondents were asked to identify the subject they find most difficult to understand without a teacher's help when studying alone at home.

Table 4.

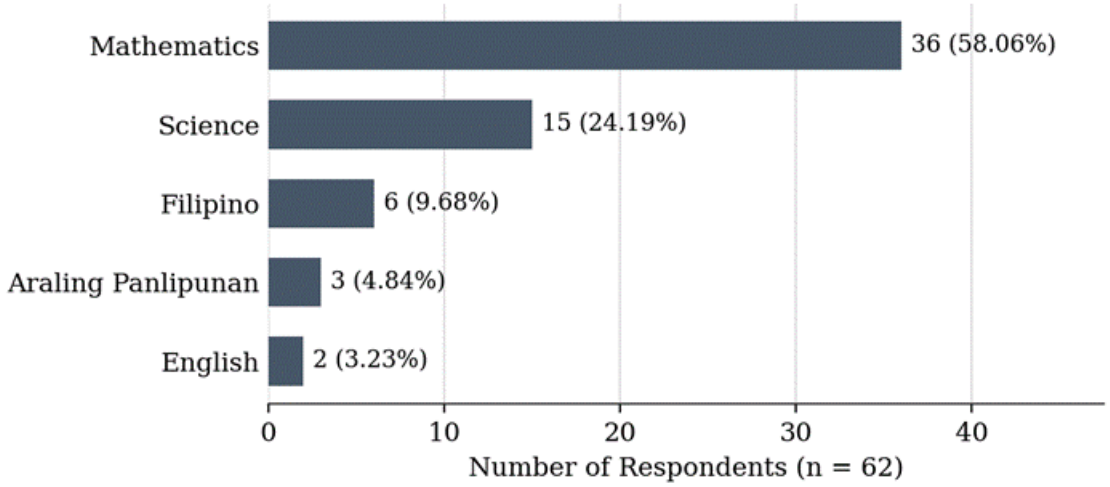
Subjects Students Find Hardest Without a Teacher

Subject Area	Frequency (f)	Percentage (%)
Mathematics	36	58.06
Science	15	24.19
Filipino	6	9.68
Araling Panlipunan	3	4.84
English	2	3.23
Total	**62**	**100.00**

Note. n = 62. The table shows which subject the students find hardest to study alone. Each student chose only one subject.

Figure 5.

Subjects Students Find Hardest Without a Teacher



Note. n = 62. The figure presents the subject results in Table 4 as a bar graph.

Mathematics was identified by 36 of the 62 respondents, or 58.06 percent, followed by Science at 24.19 percent, Filipino at 9.68 percent, Araling Panlipunan at 4.84 percent, and English at 3.23 percent. Mathematics and Science together accounted for 51 of the 62 responses, or 82.26 percent.

Table 5.

Hardest Subjects by Grade Level

Subject Area	Grade 7	Grade 8	Grade 9	Grade 10	Total
Mathematics	2	16	11	7	36
Science	5	4	3	3	15
Filipino	0	3	1	2	6
Araling Panlipunan	1	1	1	0	3

English	1	0	1	0	2
Total	**9**	**24**	**17**	**12**	**62**

Note. n = 62. The table shows the same subject choices broken down by grade level. Entries are counts of students.

Mathematics was the most frequently identified subject in Grades 8, 9, and 10, while Science was most frequently identified in Grade 7. Science was also second in Grades 8, 9, and 10, making it the only subject after Mathematics identified across all four grade levels. These results informed the selection of the subject area used as the initial learning content during the implementation phase. Science was designated for this purpose because it is compatible with the concept-based learning capabilities of LearnSync AI established in the scope of the study.

Metacognitive and Comprehension Challenges in Independent Study

Table 6 presents the responses of the sixty-two (62) respondents to the four indicators that describe the difficulties they encounter in monitoring and verifying their own understanding while studying independently.

Table 6.

Difficulty Checking One's Own Understanding

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			

3. When reviewing on my own, I find it difficult to identify exactly which specific topics or skills I need to spend more time studying.	18	25	19	0	2.98	0.78	Agree
4. When studying independently, I often think I understand a lesson, but I only realize I am wrong when I actually take the test in school.	18	17	23	4	2.79	0.94	Agree
7. I struggle to remember and understand complex lessons when I just read my physical hardbound books without anyone to explain them to me.	15	30	15	2	2.94	0.79	Agree
8. When I try to explain a difficult concept out loud to myself to see if I understand it, I have no way to check if my explanation is actually correct.	16	23	19	4	2.82	0.90	Agree
Composite Mean					**2.88**	**0.62**	**Agree**

Note. n = 62. The table shows how the students answered the four statements about checking their own understanding. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

All four indicators were interpreted as Agree. Difficulty identifying which topics require more study time obtained the highest weighted mean at 2.98, followed by difficulty understanding complex lessons from hardbound books at 2.94 and inability to verify one's own explanation at 2.82. The lowest was the tendency to believe a lesson has been understood until the test proves otherwise, at 2.79, on which 35 respondents (56.45 percent) agreed or strongly agreed and 27 (43.55 percent) disagreed or strongly disagreed. The thematic

area obtained a composite mean of 2.88 (SD = 0.62). In frequency terms, 45 respondents (72.58 percent) agreed or strongly agreed that they struggle to understand complex lessons from printed materials without explanation.

Engagement and Motivation Challenges During Independent Study

Table 7 presents the responses to the two indicators that describe the respondents' attention and motivation during independent study.

Table 7.

Difficulty Staying Engaged While Studying Alone

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
5. I lose focus quickly when my independent study relies entirely on reading traditional, plain-text materials.	21	21	14	6	2.92	0.98	Agree
6. I would study longer and more consistently at home if my review materials included game-like features.	30	18	12	2	3.23	0.88	Agree
Composite Mean					**3.07**	**0.75**	**Agree**

Note. n = 62. The table shows how the students answered the two statements about losing focus and staying motivated. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

Both indicators were interpreted as Agree. The statement that game-like features would lengthen and regularize study sessions obtained a weighted

mean of 3.23, with 30 respondents strongly agreeing and 48 (77.42 percent) agreeing or strongly agreeing. Losing focus when studying from plain-text materials obtained 2.92, with 42 respondents (67.74 percent) agreeing or strongly agreeing. The thematic area obtained a composite mean of 3.07 (SD = 0.75).

Limitations of Currently Used Digital Study Tools

Table 8 presents the responses to the four indicators that describe the limitations the respondents encounter in the educational applications and websites they currently use for independent study.

Table 8.

Limits of the Study Apps Students Already Use

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
10. Most educational apps I use just give me a final score but do not automatically adapt to my weaknesses or tell me what specific topics I need to study next.	17	30	11	4	2.97	0.85	Agree
11. The study apps I currently use rely heavily on multiple-choice questions, which makes it easy to guess	18	20	20	4	2.84	0.93	Agree

the answer without actually understanding the lesson.							
12. I find it frustrating that current educational apps do not allow me to type out or explain my answers in my own words to check if my reasoning is correct.	18	27	11	6	2.92	0.93	Agree
13. I have to use too many different apps or websites because there is no single app that does everything I need for independent study.	25	15	18	4	2.98	0.98	Agree
Composite Mean					**2.93**	**0.65**	**Agree**

Note. n = 62. The table shows how the students answered the four statements about the study apps they already use. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

All four indicators were interpreted as Agree. Having to use several different applications obtained the highest weighted mean at 2.98, followed by applications providing only a final score without adapting to weaknesses at 2.97 and the inability to type or explain answers at 2.92. Heavy reliance on multiple-choice questions, which allows answers to be guessed, obtained 2.84, on which 38 respondents (61.29 percent) agreed or strongly agreed and 24 (38.71 percent) disagreed or strongly disagreed. The thematic area obtained a composite mean of 2.93 (SD = 0.65). Forty-seven respondents (75.81 percent) agreed or strongly agreed that their applications do not adapt to their weaknesses.

Expressed Need for an Integrated and Adaptive Learning Platform

Table 9 presents the responses to the two indicators that describe the respondents' expressed need for a single learning platform that tracks progress and consolidates their independent study requirements.

Table 9.

Need for One Adaptive Study Platform

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
9. I would study more consistently at home if I had an interactive, digital tool that could track my progress and tell me exactly where I need to improve.	31	23	6	2	3.34	0.79	Strongly Agree
14. Having a single, all-in-one application that contains everything I need for independent learning would significantly improve my study habits.	35	18	9	0	3.42	0.74	Strongly Agree
Composite Mean					**3.38**	**0.61**	**Strongly Agree**

Note. n = 62. The table shows how the students answered the two statements about wanting one app that tracks their progress. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

Both indicators were interpreted as Strongly Agree. The statement that a single all-in-one application would significantly improve study habits obtained the

highest weighted mean of the entire instrument at 3.42, with 35 respondents strongly agreeing and none strongly disagreeing. The need for an interactive tool that tracks progress and indicates where to improve obtained 3.34. The thematic area obtained a composite mean of 3.38 (SD = 0.61). Fifty-four respondents (87.10 percent) agreed or strongly agreed with the need for progress tracking, and 53 (85.48 percent) with the need for an all-in-one application.

Summary of the Requirements Analysis Findings

Table 10 presents the composite means of the four thematic areas covered by the requirements analysis questionnaire, and Figure 6 presents the weighted mean of each of the twelve indicators grouped according to these thematic areas.

Table 10.

Summary of Survey Findings by Thematic Area

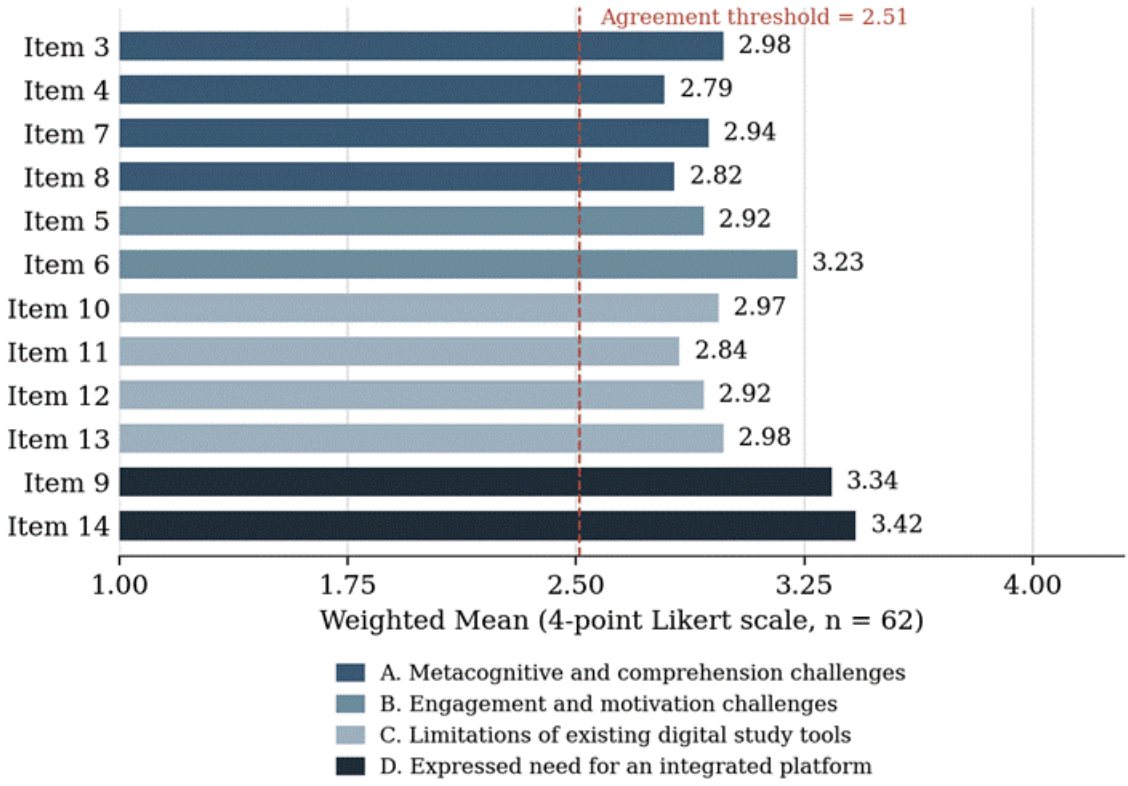
Thematic Area	No. of Items	Composite Mean	Standard Deviation	Verbal Interpretation	Rank
A. Metacognitive and comprehension challenges in independent study	4	2.88	0.62	Agree	4
B. Engagement and motivation challenges during independent study	2	3.07	0.75	Agree	2

C. Limitations of currently used digital study tools	4	2.93	0.65	Agree	3
D. Expressed need for an integrated and adaptive learning platform	2	3.38	0.61	Strongly Agree	1
Overall Composite Mean	**12**	**3.01**	**0.45**	**Agree**	

Note. n = 62. The table summarizes the four groups of statements and ranks them by their composite mean.

Figure 6.

Survey Results by Thematic Area



Note. n = 62. The figure shows the weighted mean of each of the twelve statements, grouped by thematic area. The dashed line marks 2.51, the point where a score counts as agreement.

The expressed need for an integrated and adaptive platform ranked first at 3.38, interpreted as Strongly Agree, followed by engagement and motivation challenges at 3.07, limitations of existing tools at 2.93, and metacognitive and comprehension challenges at 2.88, all interpreted as Agree. The overall composite mean was 3.01 (SD = 0.45). All twelve indicators exceeded the agreement threshold of 2.51, and none fell within the Disagree or Strongly Disagree ranges.

Discussion

The findings show that independent learning at Pasig Christian Academy is primarily text-based and unsupervised, and that the reported difficulties concentrate on monitoring one's own understanding rather than on access to materials. Respondents agreed that they cannot reliably identify which topics need more study, cannot verify their own explanations, and often discover misconceptions only during classroom testing. These results align with literature reporting that learners differ in their ability to set goals, monitor progress, and seek help during independent study, and that weaker self-regulation is associated with poorer outcomes (Briones et al., 2023; Calamlam, 2023; Funa et al., 2023), and with reports that Filipino learners struggle with self-study modules where supervision is reduced (Abejuela et al., 2023; Agayon et al., 2022).

The identification of Mathematics and Science by 82.26 percent of respondents indicates that difficulty concentrates in subjects requiring conceptual

explanation and procedural reasoning rather than recall alone. Agreement that plain-text materials cause loss of focus and that game-like review would lengthen study sessions corresponds with literature reporting that gamified activities support attention, motivation, and sustained engagement (Al-Khresheh, 2025; Dahalan et al., 2023). The reported inability to verify one's own explanation corresponds with work on explanation-based and retrieval-oriented learning, which finds that explaining material in one's own words supports comprehension but requires feedback on accuracy to be productive (Corral & Carpenter, 2025; Fiorella, 2023; Serra et al., 2025).

The reported limitations of existing tools, a final score without adaptation, multiple-choice items that permit guessing, and no way to answer in one's own words, parallel the literature contrasting static review systems with adaptive systems that analyze performance to identify weaknesses in real time (Akintola et al., 2025; Lee & Kim, 2024; Onesi-Ozigagun et al., 2024). The concern about guessing is also addressed in knowledge tracing, where probabilistic models estimate knowledge while accounting for guessing and slipping (Bulut et al., 2023).

The highest-rated area was the expressed need for an integrated platform, with 87.10 percent agreeing on progress tracking and 85.48 percent on an all-in-one application. This is supported by literature on learning analytics dashboards, which reports that making progress visible improves awareness, self-efficacy, and engagement when designed to support regulation rather than merely display data (Fleur et al., 2023; Paulsen & Lindsay, 2024; Yang et al.,

2024), and by studies indicating that integrated adaptive platforms suit learners still developing independent study habits (Barrera Castro et al., 2024; Caliston, 2025; Salva et al., 2022).

Taken together, the findings establish the need for a platform that estimates and displays topic-level mastery, evaluates explanations expressed in the learner's own words, sustains engagement through gamified review, makes progress continuously visible, and consolidates these functions in one place. These findings were consolidated into the system requirements presented under Statement of the Problem No. 2.

FEATURES AND FUNCTIONALITIES INCORPORATED INTO LEARNSYNC AI

This section presents the features incorporated into LearnSync AI in response to the challenges established under Statement of the Problem No. 1. The requirements analysis established the problems students encounter rather than the features to be built. The findings were therefore translated into system requirements, and the method implemented for each requirement was selected from the approaches reviewed in Chapter I rather than proposed by the respondents.

Derived System Requirements

Table 11 presents the system requirements derived from the requirements analysis findings, together with the indicators that support each requirement and the thematic area under which those indicators were reported in Tables 6 to 9.

Table 11.

System Requirements Drawn from the Survey

Req. ID	Derived System Requirement	Supporting Indicators (Weighted Mean)	Thematic Area
R1	Estimation and display of topic-level knowledge so that students can identify which concepts require further study	Items 3 (2.98), 4 (2.79), 10 (2.97)	A, C
R2	Assessment that accepts answers expressed in the student's own words rather than selection alone	Items 11 (2.84), 12 (2.92)	C
R3	Evaluation of the correctness of a student's own explanation of a concept	Items 7 (2.94), 8 (2.82)	A
R4	Gamified and interactive review activities calibrated to the learner	Items 5 (2.92), 6 (3.23)	B
R5	Visible and continuous tracking of learning progress with direction on where to improve	Item 9 (3.34)	D
R6	Consolidation of independent study functions within a single platform	Items 13 (2.98), 14 (3.42)	C, D

R7	Educator-managed learning content with grade-level access control and class-level knowledge monitoring	Requirements analysis interviews	Educator

Note. The table lists the seven system requirements drawn from the survey findings, together with the survey items that support each one. R7 came from the interviews with the principal and the educator, so it has no student item.

Seven requirements were derived. R1 to R6 came from the student questionnaire, while R7 came from the interviews conducted with the school principal and the educator during the same pre-development phase. The requirement with the strongest supporting indicators was R6, drawing from Item 14 at 3.42 and Item 13 at 2.98, followed by R5 from Item 9 at 3.34.

Requirements-to-Features Traceability

Table 12 presents the traceability matrix mapping each feature and functionality of LearnSync AI to the requirements it addresses. The matrix was used during the analysis and design phase to verify that every derived requirement was addressed by at least one implemented feature and that no feature was developed without a corresponding requirement.

Table 12.

Requirements-to-Features Traceability Matrix

Feature / Functionality	R1	R2	R3	R4	R5	R6	R7

Adaptive Flashcard Mode	✓					✓	
Adaptive Test Mode	✓	✓				✓	
Free Recall with AI Evaluation		✓	✓			✓	
Feynman Technique Module		✓	✓			✓	
AI Learning Assistant			✓			✓	
PlayMe Gamification Portal				✓		✓	
Student Dashboard and Learning Analytics	✓				✓	✓	
Study Space Hub						✓	
Educator Portal					✓	✓	✓

Note. A check mark means the feature answers that requirement. The table shows that every requirement is answered by at least one feature, and that every feature answers at least one requirement.

All seven requirements are addressed, and no feature was developed without a traceable requirement. R1 is addressed by Adaptive Flashcard Mode, Adaptive Test Mode, and the Student Dashboard, the modules through which the BKT engine updates and displays estimates. R2 is addressed by the identification and fill-in-the-blank formats of Adaptive Test Mode and by Free Recall and the Feynman Technique, which accept free-form input. R3 is addressed by Free Recall, the Feynman Technique, and the AI Learning Assistant, which evaluate or respond to student explanations. R4 is addressed by

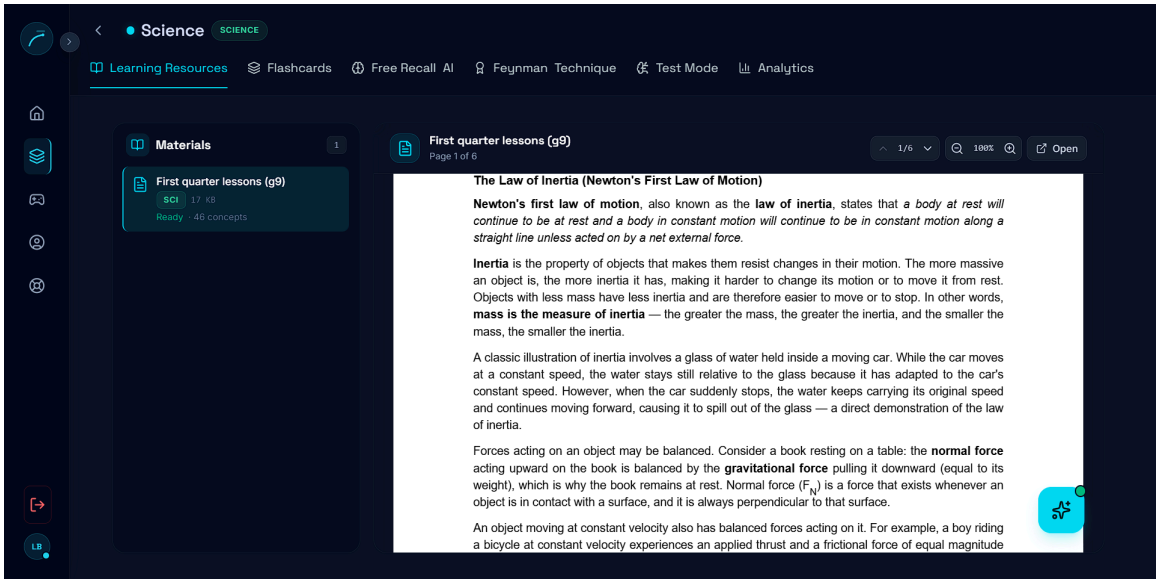
the PlayMe Gamification Portal, R5 by the Student Dashboard and Educator Portal analytics, and R7 by the Educator Portal.

Student Portal Features

The Study Space Hub is the single point of access through which students open materials and select a study mode. Consolidating the modes in one interface addresses R6, which arose from the finding that 64.52 percent of respondents agreed or strongly agreed that they must use several different applications for independent study and 85.48 percent agreed an all-in-one application would improve their habits.

Figure 7.

Study Space Hub

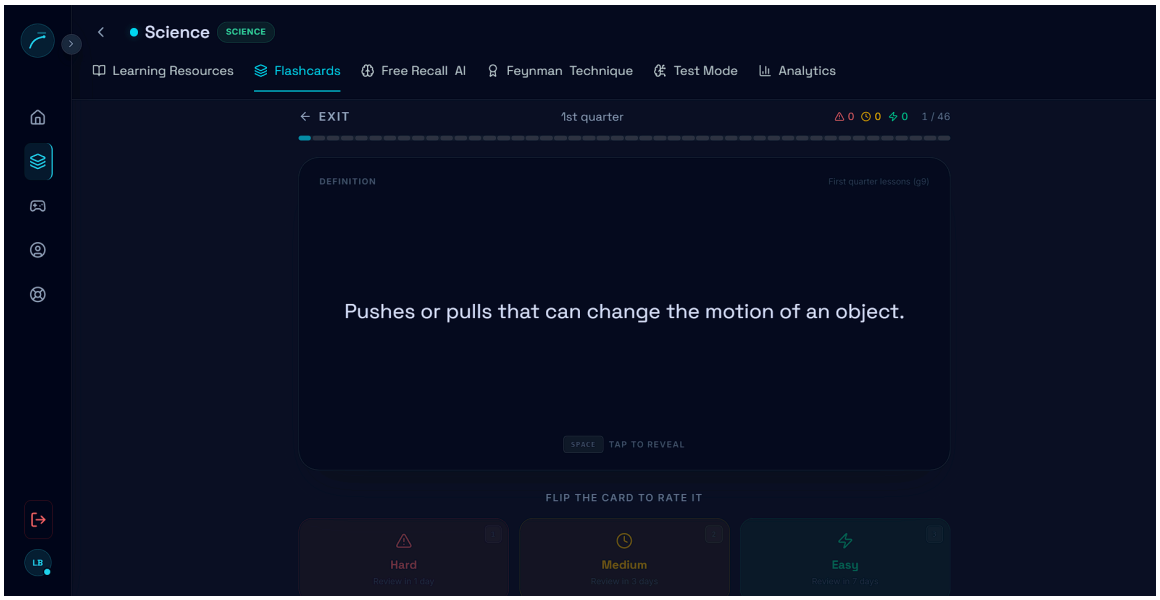


Note. The figure shows the learning resources page of the application wherein the students can view and read the uploaded material of the educator.

Adaptive Flashcard Mode presents concept-level flashcards extracted from the uploaded material and lets the student rate each card as Easy, Medium, or Hard. Every rating is a practice opportunity, and the BKT engine updates the knowledge probability for that concept. Adaptive Test Mode lets the student select from five question formats, grades responses as binary values, and updates the estimate per question; when a new test is requested on the same topic, the stored probability scales the difficulty of the generated items. Together these modes address R1 and, through the open-response formats, R2.

Figure 8.

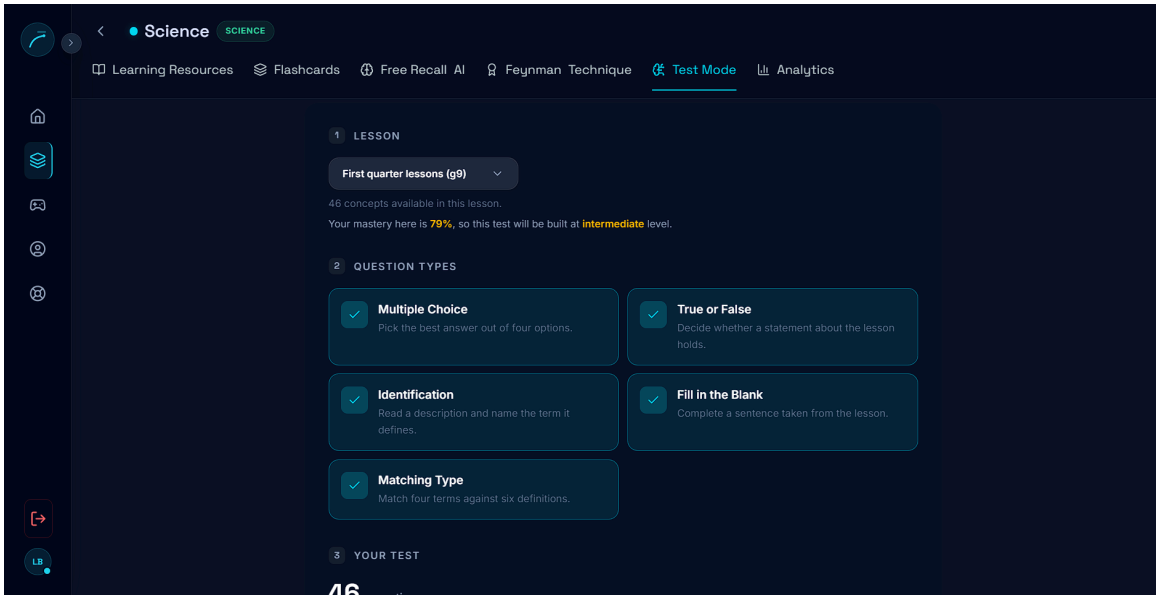
Adaptive Flashcard Mode



Note. The figure shows the flashcard page of the application wherein the students can review the learning materials using flashcards and rate the difficulty of each card.

Figure 9.

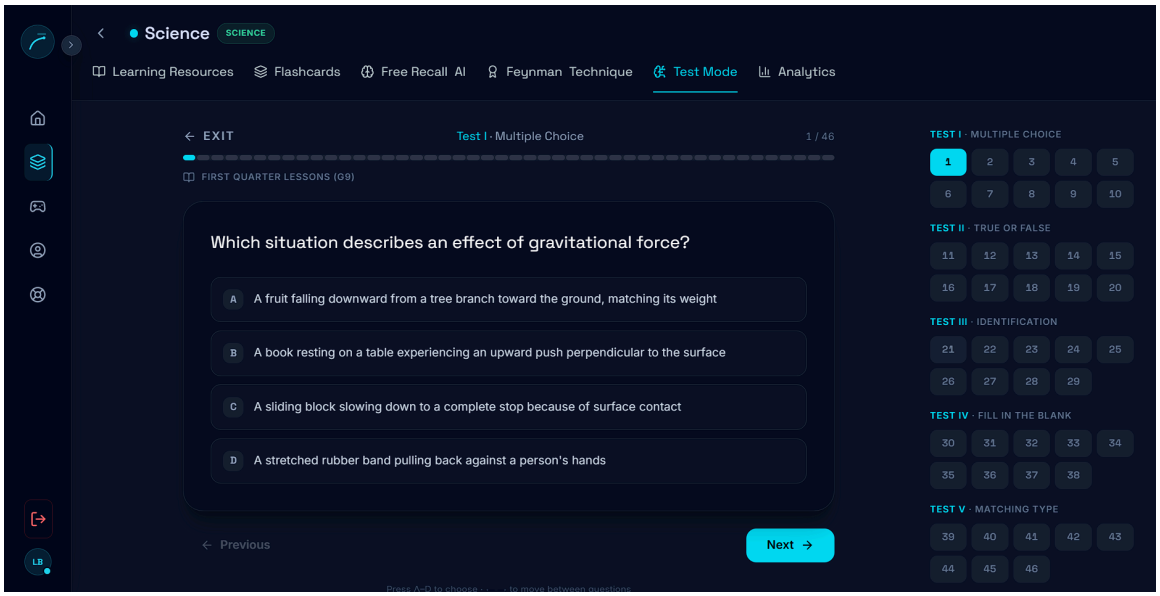
Adaptive Test Mode: Choosing a Format



Note. The figure shows the test mode page of the application wherein the students can select which mode of questions will the generated test contain.

Figure 10.

Adaptive Test Mode: Taking the Test



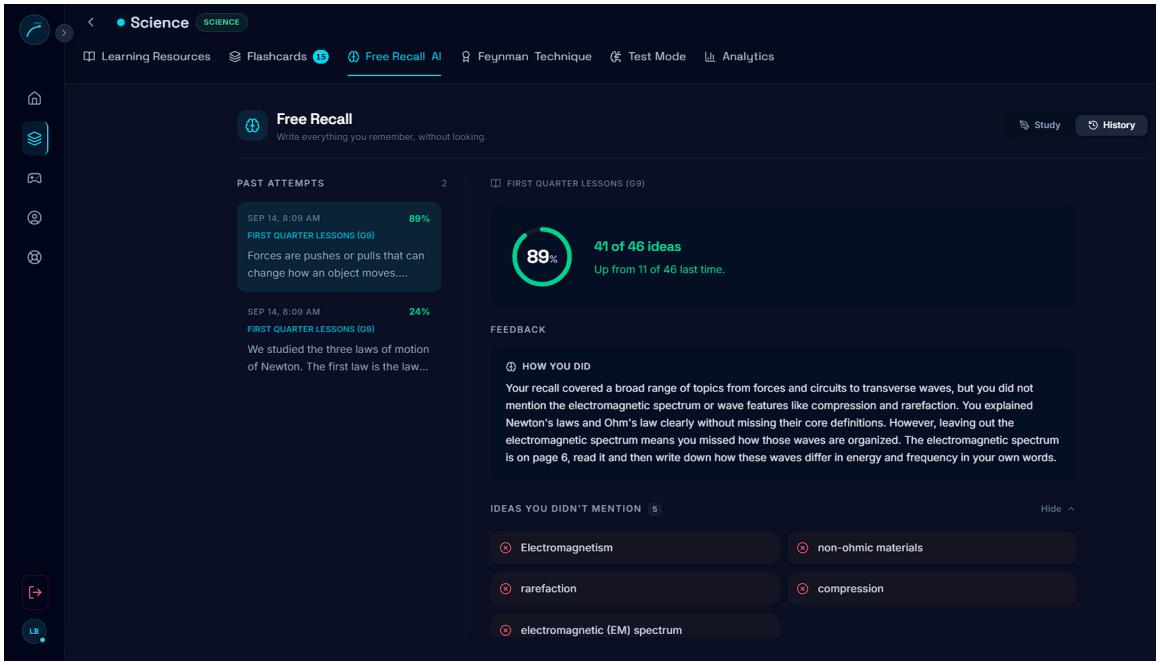
Note. The figure shows the test mode page of the application wherein the students can now answer the generated test that the system produces.

The Free Recall module lets students type everything they remember about a lesson without referring to notes. The response is converted into a vector embedding and compared against the stored lesson vectors using cosine similarity computed in the backend. When the score falls below the 0.70 threshold, the system compares the response against the lesson glossary and identifies the missing keywords. The Feynman Technique module asks students to explain a concept in simple language and evaluates it in a single request

returning one structured result covering three checks: Jargon Check, Analogy Verification, and Gap Finding, producing an accuracy score from 0 to 100, a verdict, and written feedback. These modules address R3, which arose from the finding that 62.90 percent of respondents agreed or strongly agreed that they have no way to check whether their own spoken explanation of a concept is correct, and R2, from the finding that 72.58 percent agreed or strongly agreed that current applications do not allow them to explain answers in their own words.

Figure 11.

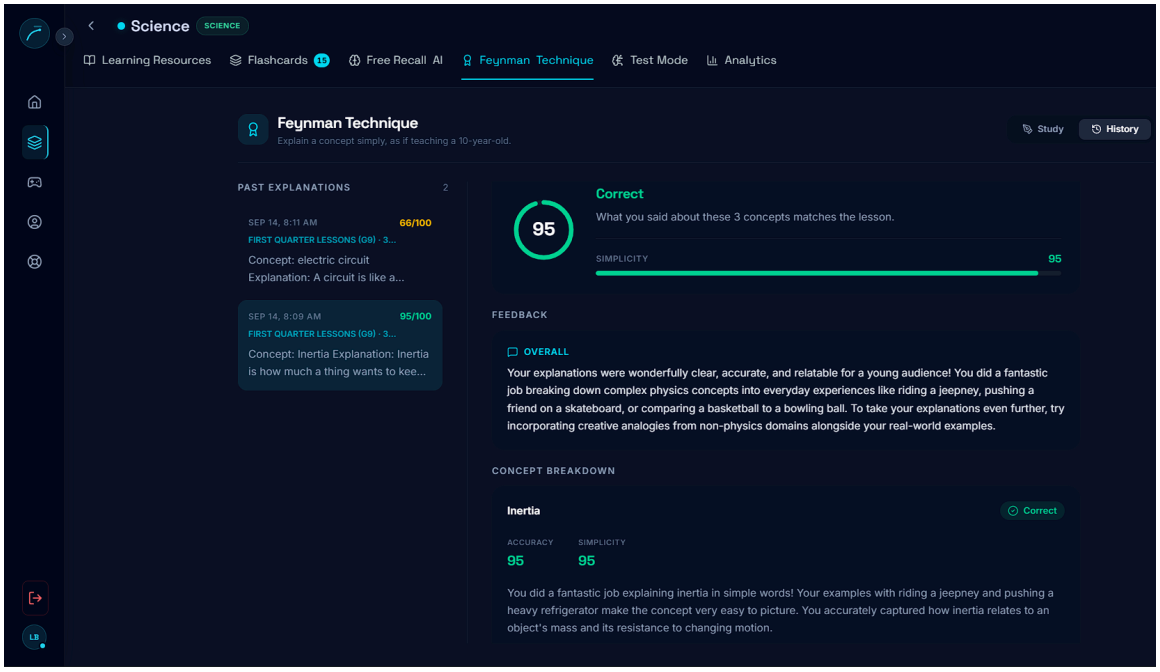
Free Recall Module



Note. The figure shows the Free recall page of the application showing the result of what the student recalled.

Figure 12.

Feynman Technique Module

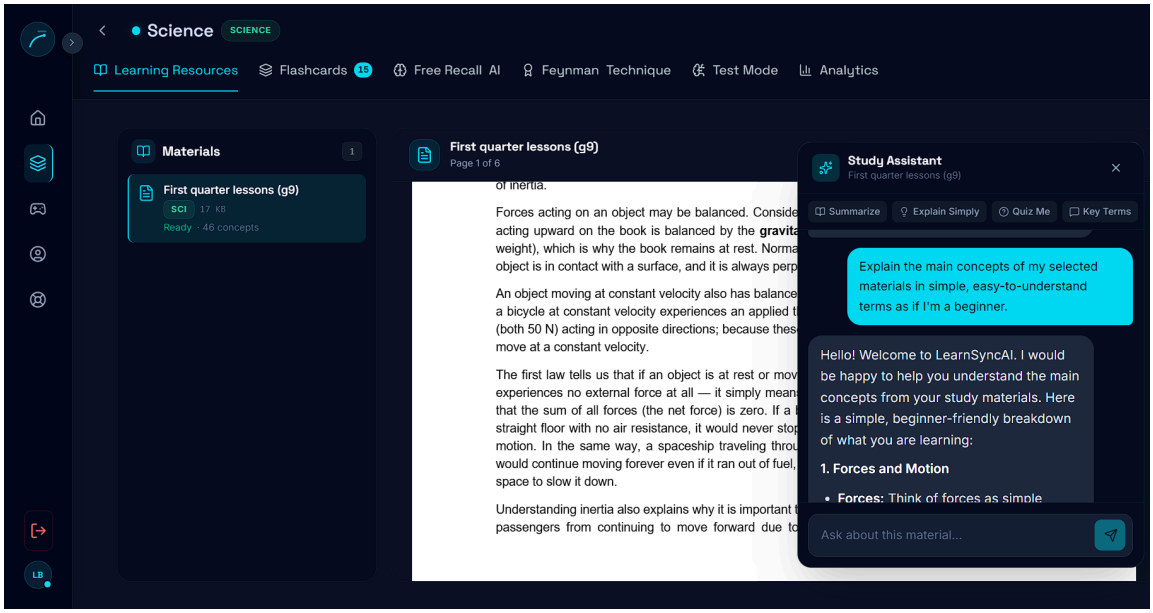


Note. The figure shows the Feynman technique page of the application showing the result of the student's simplified explanation of the selected concept.

The AI Learning Assistant answers student questions with responses grounded in the educator-uploaded resource through vector retrieval, addressing R3 by providing explanations on demand. The PlayMe Gamification Portal provides Crossword, Term Matching, and Timed Trivia, whose vocabulary complexity, clue difficulty, and response timers are constrained by the student's document knowledge index. This addresses R4, which arose from the finding that 77.42 percent of respondents agreed game-like review would lengthen and regularize their study.

Figure 13.

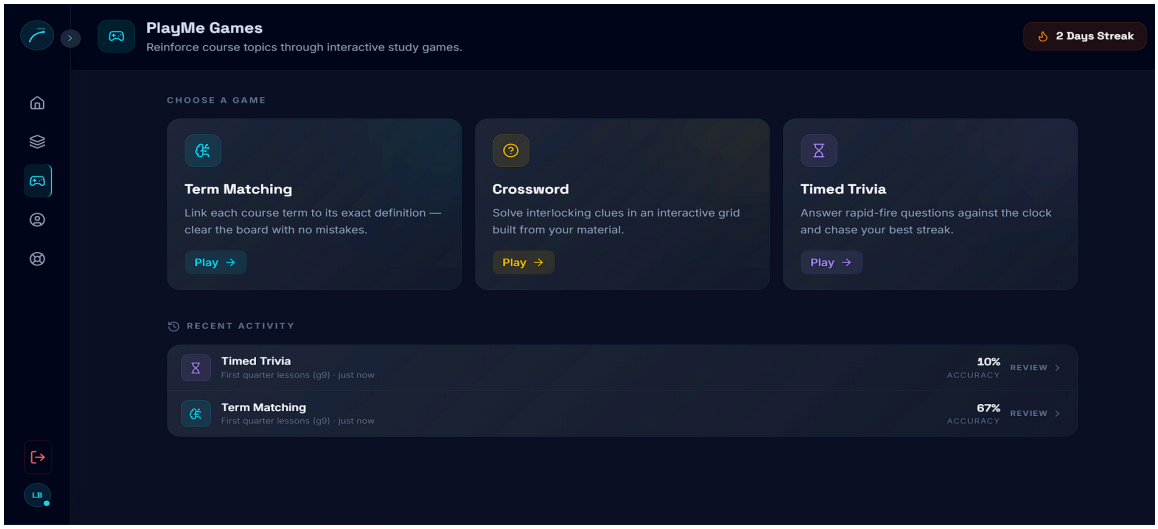
AI Learning Assistant



Note. The figure shows the Learning Resource page of the application showing the assistant's reply to a student's question

Figure 14.

PlayMe Gamification Portal

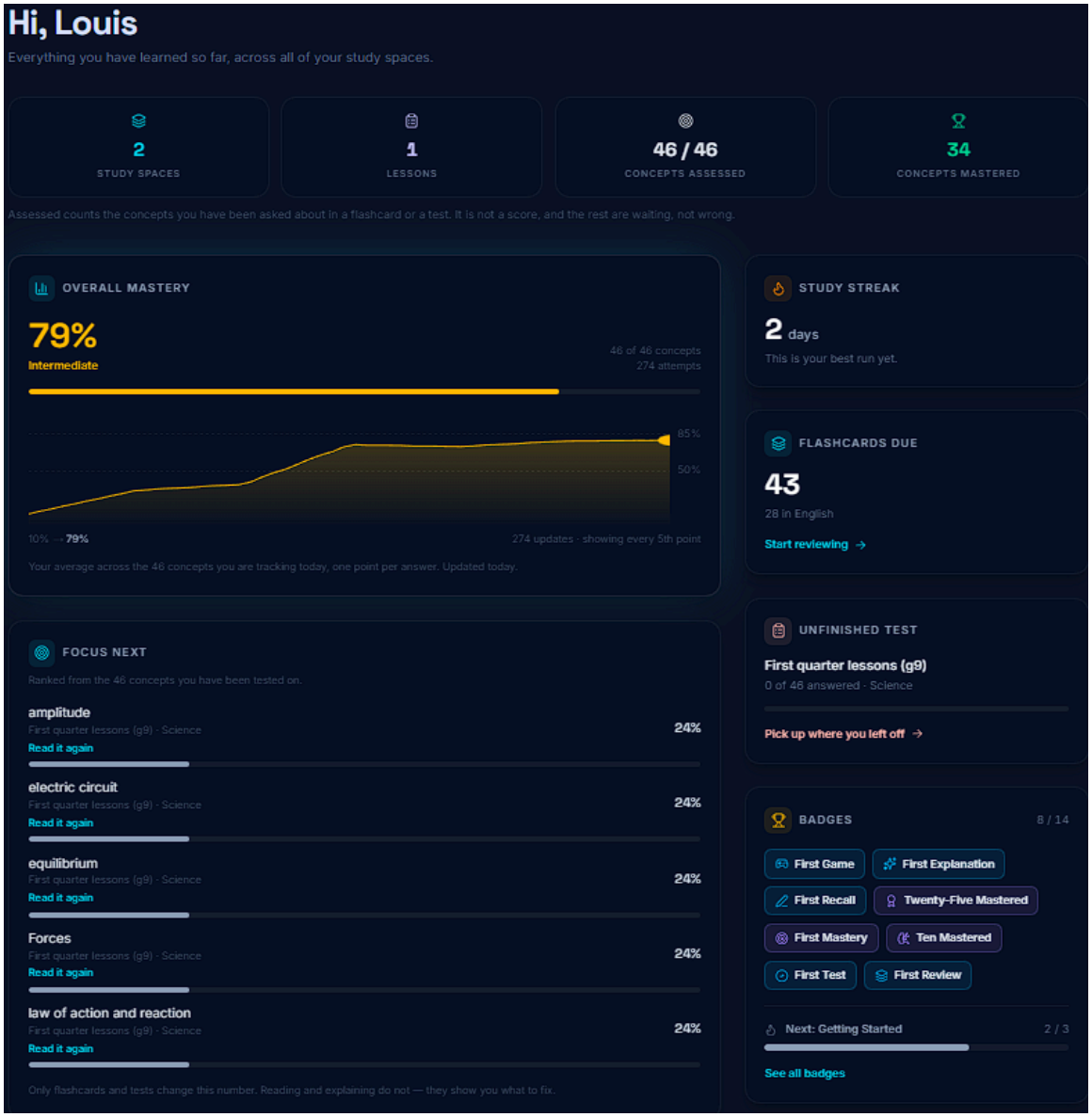


Note. The figure shows the Games page of the application wherein it shows the available games that the application offers.

The Student Dashboard displays the document knowledge index as a progress bar, knowledge curves across practice opportunities, daily streaks, and earned badges. This addresses R5, which arose from the finding that 87.10 percent of respondents agreed an interactive tool tracking progress would make their study more consistent, and supports R1 by making topic-level estimates visible rather than keeping them in the database.

Figure 15.

Student Dashboard



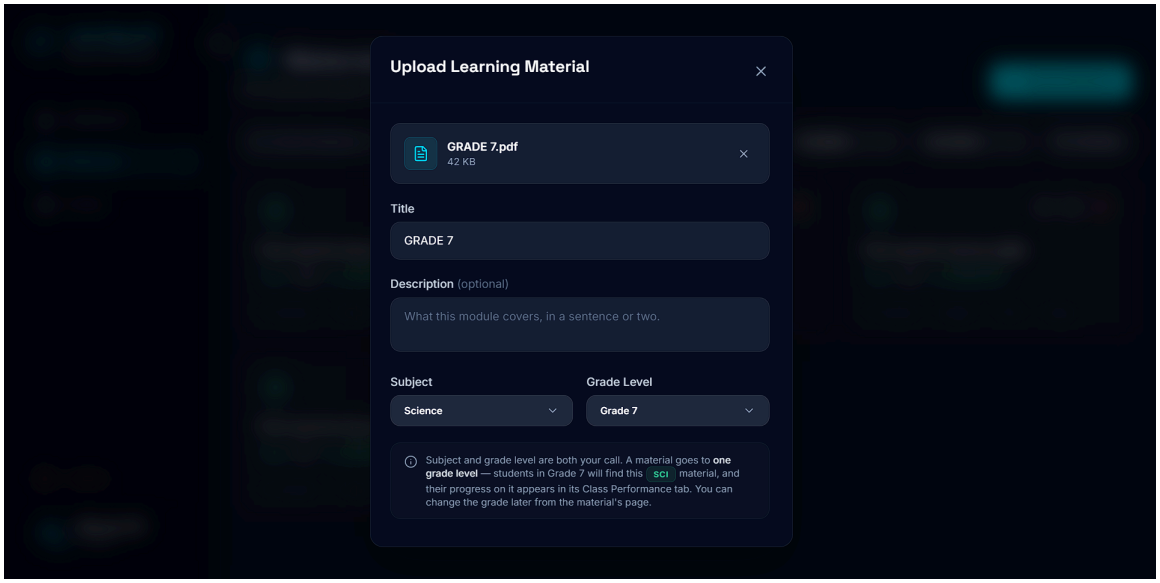
Note. The figure shows the Dashboard page of the student wherein it shows their overall mastery, concepts that they need to focus on, the badges they have earned, and their study streaks

Educator Portal Features

The Educator Portal allows the educator to upload materials as plain-text PDF modules or images of printed pages, to select which grade level may access each material, and to view a real-time analytics overview of student knowledge aggregated at the concept and document level. Image uploads pass through Optical Character Recognition before entering the same parsing pipeline as digital modules, which allows lessons existing only in printed textbooks to be used. These functions address R7 and contribute to R5 by giving the educator a view of class-level progress.

Figure 16.

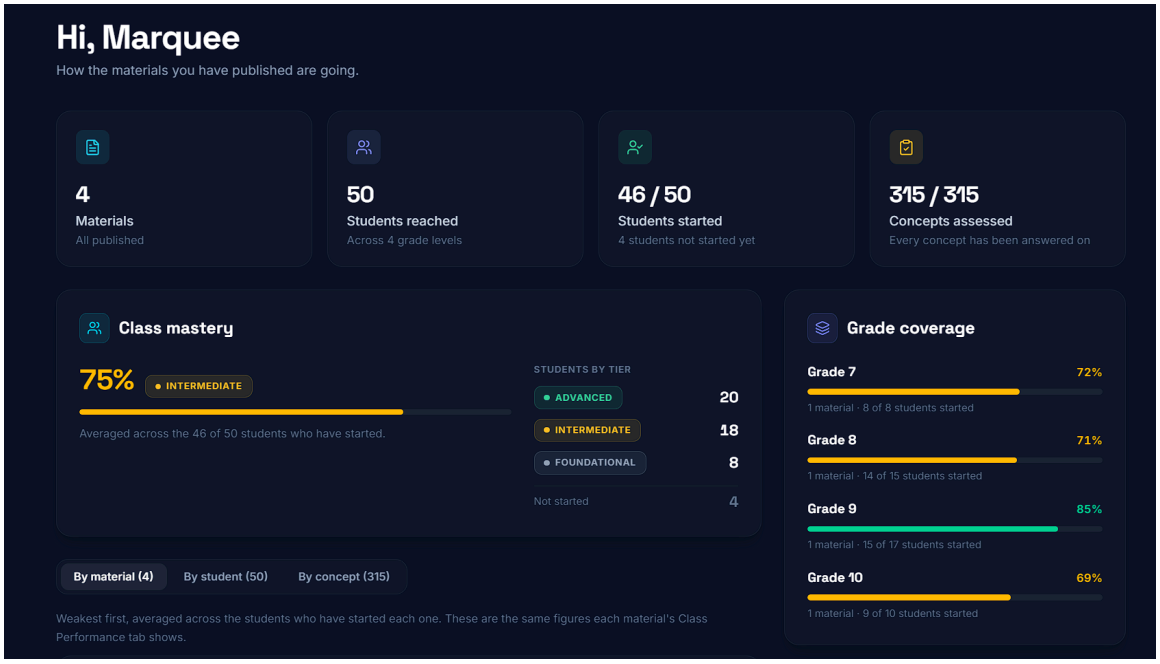
Educator Portal: Uploading Material



Note. The figure shows the upload page wherein the educators can upload their learning materials that the students can use.

Figure 17.

Educator Dashboard



Note. The figure shows the Dashboard page of the educator user wherein it shows the knowledge percentage of their students.

Integration of Bayesian Knowledge Tracing

The classical four-parameter BKT model was integrated as the cognitive modeling core. Each extracted concept is initialized at a baseline prior of 0.10 and updated after every practice opportunity recorded in Adaptive Flashcard Mode and Adaptive Test Mode. Rather than applying one fixed guess probability, the value was calibrated to the response format, since the baseline chance of answering correctly without knowledge differs between a two-option item and an open-ended one.

Table 13.

Guess Probability by Assessment Mode

Assessment Mode	Response Format or Rating	Guess Probability P(G)	Basis for the Value
Adaptive Flashcard Mode	Easy	0.05	High confidence that the correct response reflected genuine knowledge
Adaptive Flashcard Mode	Medium	0.25	Moderate possibility that the response was aided by partial guessing
Adaptive Flashcard Mode	Hard	0.10	Recorded as an incorrect response; produces a steeper decrease in knowledge
Adaptive Test Mode	Multiple Choice (4 options)	0.25	Baseline probability of a correct random response of 1/4
Adaptive Test Mode	True or False	0.50	Baseline probability of a correct random response of 1/2
Adaptive Test Mode	Matching Type (4 terms, 6 definitions)	0.17	Baseline probability of 1/6 applied per matched item
Adaptive Test Mode	Identification	0.05	No provided options; random production of the exact term is negligible
Adaptive Test Mode	Fill in the Blank	0.05	No provided options; free recall of the term without cues

Note. The table shows the guess value used for each answer format. A lower value means a correct answer raises the knowledge estimate more. $P(L_0) = 0.10$ and $P(S) = 0.15$.

The guess probability ranges from 0.05 for identification and fill-in-the-blank to 0.50 for true or false. Open-response formats were assigned the lowest value because they provide no options, so a correct response contributes the largest increase; true or false was assigned the highest, so a correct binary-choice response contributes substantially less. This configuration directly addresses R2, since it makes the measured knowledge gain depend on whether the student produced the answer or selected it. The aggregated document knowledge index, the arithmetic mean of current probabilities across all concepts in a document, is displayed to the student as a progress indicator and passed to the gamification engine to constrain game difficulty.

Integration of Generative Artificial Intelligence

Generative Artificial Intelligence was integrated across seven components of the platform, spanning content ingestion, assessment generation, response evaluation, and gamified content production. Table 14 presents each component together with its function and the output it produces.

Table 14.

Generative AI Components and Their Functions

Generative AI Component	Function	Output
Document parsing and concept extraction	Extracts key terms, concepts, and definitions from uploaded materials and generates vector embeddings	Concept nodes and glossary used by all study modes

Optical Character Recognition	Converts images of printed textbook pages into machine-readable text	Text that enters the same parsing pipeline as digital modules
Adaptive test item generation	Generates practice items in five formats, scaled by the running BKT knowledge probability	Advanced tier at P(L) of 0.85 and above, intermediate at 0.50 to 0.85, foundational below 0.50
Free Recall evaluation	Compares the student's typed recall against the stored lesson vectors and glossary	Similarity score and a list of missing keywords, threshold 0.70
Feynman Technique evaluation	Applies one structured evaluation covering Jargon Check, Analogy Verification, and Gap Finding	Accuracy score of 0 to 100 and a verdict of Correct, Needs Improvement, or Incorrect
AI Learning Assistant	Answers student questions through vector retrieval from the uploaded resource	Responses grounded in the educator's material
Procedural game generation	Generates crosswords, term matches, and trivia constrained by the document knowledge index	Game content calibrated to the student's knowledge level

Note. The table lists the seven parts of the system that use Generative AI and what each one produces. All of them work only on the materials the educator uploaded.

The components operate as a connected pipeline rather than independent utilities. The parsing component produces the concept nodes and glossary on which every other component depends: the test generator draws items from those concepts, the Free Recall evaluator compares against the stored vectors and glossary, the Feynman evaluator checks jargon against the same glossary, the AI Learning Assistant retrieves from the same embeddings, and the game generator draws vocabulary from the same parsed content. Every activity presented to the student therefore derives from the material the educator uploaded rather than from general model knowledge.

Discussion

The features were determined by the requirements established in the preceding section rather than selected independently of the identified needs. The traceability matrix in Table 12 shows complete bidirectional coverage, with every requirement addressed by at least one feature and every feature tracing to at least one requirement. This correspondence is consistent with the developmental research design described in Chapter II, in which educational needs are translated systematically into functional features.

Selecting Bayesian Knowledge Tracing as the cognitive core responds to the finding that students cannot identify which topics require further study. The model estimates knowledge per concept while accounting for guessing and slipping, and remains interpretable enough to display to a learner as a mastery estimate rather than a score, properties consistent with the literature describing BKT as a widely used and interpretable approach to learner modeling (Abdelrahman et al., 2023; Bulut et al., 2023). Varying the guess parameter by response format follows evidence that allowing guess and slip to vary between responses produces more accurate learner models than fixed per-concept values (Šarić-Grgić et al., 2024).

Using Generative AI for adaptive item generation responds to the finding that existing applications report a score without adapting to weaknesses. Conditioning item difficulty on the running knowledge estimate corresponds with literature on adaptive systems that analyze performance to recommend targeted

activities (Akintola et al., 2025) and with reported implementations of AI-driven assessment generation (Al-Gburi, 2025; Bahroun et al., 2023). Grounding the AI Learning Assistant in educator-uploaded resources rather than general model knowledge corresponds with approaches that constrain generative responses to an authoritative source to maintain factual alignment (Sajja et al., 2023).

The Free Recall and Feynman Technique modules operationalize two strategies the literature identifies as effective but that students reported being unable to verify alone. Retrieval-based study is reported to be more effective than passive review for retention and metacognitive monitoring (Corral & Carpenter, 2025; Serra et al., 2025), and explanation-based tasks are reported to support comprehension and knowledge generalization when the learner receives feedback on the accuracy of the explanation (Fiorella, 2023). The automated evaluation supplies the feedback component that the respondents identified as missing.

The PlayMe Portal responds to the finding that students lose focus with plain-text materials. Calibrating game difficulty to the current knowledge index is consistent with reports that gamified activities support attention, motivation, and sustained engagement (Al-Khresheh, 2025; Dahalan et al., 2023; Pellas, 2024), including crossword-type activities in structured review (Arnold et al., 2024). The Student Dashboard responds to the highest-rated requirement by making progress visible, an approach reported to improve learners' awareness and engagement when designed to support regulation rather than merely display data (Fleur et al., 2023; Paulsen & Lindsay, 2024; Yang et al., 2024). These features

constitute the developed system whose correctness, recorded learning progress, and acceptability are reported under Statements of the Problem No. 3, No. 4, and No. 5.

VERIFICATION OF THE BAYESIAN KNOWLEDGE TRACING ENGINE

This section presents the verification conducted on the BKT engine. Following the deterministic algorithm verification described in Chapter II, the procedure determined whether the knowledge estimates generated by the deployed system are mathematically identical to values obtained when the same equations are applied outside the application to the same responses. The verification had three parts: 46 designed test cases executed in the live application (Part A), 40 student–concept sequences drawn at random from implementation records (Part B), and the complete population of recorded updates (Part C).

Verification Procedure

The engine maintains one knowledge probability for each student and concept, updating it after every practice opportunity using EQ (1) or EQ (2) followed by EQ (3). The parameters used were $P(L_0) = 0.10$, $P(T) = 0.20$, and $P(S) = 0.15$, with the context-sensitive guess probabilities in Table 13, under which Easy and Medium flashcard ratings enter as correct observations and

Hard enters as incorrect. One test case is one update, and a case passed when the absolute difference between system and manual values fell within the ±0.01 tolerance established in Chapter II. Correctness was quantified using EQ (6).

Expected values were computed in a companion spreadsheet in which every value is a visible formula. Only two categories of information were taken from the system: the graded observation sequence and the parameter table. Each sequence was recomputed forward from 0.10, so the expected prior at any step was the expected of the preceding step. The priors stored by the application were never used as inputs but treated as additional values to verify, which prevents an engine containing a defect from producing apparent agreement by supplying its own intermediate values. System values were read from the audit table, which records one row per update, and from the table storing the current estimate and accumulated opportunity count.

The eight guess values and the binary observation produce thirteen distinct update branches. Table 15 presents the number of updates falling under each branch in the three parts.

Table 15.

Coverage of the Thirteen Update Types

Update Branch	P(G)	Part A	Part B	Part C
Multiple choice · correct	0.25	23	27	1,924
Multiple choice · incorrect	0.25	23	16	1,144
True or false · correct	0.50	12	24	2,153

True or false · incorrect	0.50	34	4	866
Identification · correct	0.05	34	9	661
Identification · incorrect	0.05	12	2	404
Fill in the blank · correct	0.05	23	0	149
Fill in the blank · incorrect	0.05	23	0	114
Matching type · correct	0.17	22	9	368
Matching type · incorrect	0.17	22	4	288
Flashcard Easy · correct	0.05	16	21	1,494
Flashcard Medium · correct	0.25	15	19	1,328
Flashcard Hard · incorrect	0.10	15	23	1,174
Total		**274**	**158**	**12,067**

Note. The table shows how many updates of each type were checked in each part. Part A covered every branch by design; Parts B and C show how often each occurred in ordinary use.

Part A: Designed Test Cases Executed in the Live Application

Part A used a researchers' test account on a material containing 46 extracted concepts. Five adaptive tests were taken, each restricted to one question format, so every concept was answered once in each of the five formats; matching type covers four concepts per block and therefore reached 44 of the 46. Responses were chosen by design according to concept position, producing consecutive correct, consecutive incorrect, alternating, and mixed sequences, after which a flashcard deck from the same material was rated once per card. The application graded all 228 responses exactly as intended, producing 274 updates.

Table 16.

Step-by-Step Check of One Test Case

Step	Format or Rating	Outcome	P(G)	Prior	P(L | obs) EQ (1) / EQ (2)	Expected Posterior EQ (3)	System Value	Difference
1	Multiple choice	Correct	0.25	0.1000	0.2742	0.419355	0.419355	0.0000
2	True or false	Incorrect	0.50	0.4194	0.1781	0.342466	0.342466	0.0000
3	Identification	Correct	0.05	0.3425	0.8985	0.918816	0.918816	0.0000
4	Fill in the blank	Incorrect	0.05	0.9188	0.6412	0.712954	0.712954	0.0000
5	Matching type	Correct	0.17	0.7130	0.9255	0.940382	0.940382	0.0000
6	Flashcard Medium	Correct	0.25	0.9404	0.9817	0.985356	0.985356	0.0000

Note. The prior at step 1 is $P(L_0) = 0.10$. At each later step the prior is the previous step's expected posterior, computed independently of the value stored by the application.

The current estimate stored for this concept was 0.985356 after six practice opportunities, equal to the last expected posterior.

Table 17.

Results of the Forty-Six Designed Test Cases

Designed Sequence	Rating	Concepts	Steps	Updates	Expected Final	System Final	Difference	Result
All correct	Easy	11	6	66	1.0000	1.0000	0.0000	PASS

All correct	Easy	1	5	5	0.9999	0.9999	0.0000	PASS
All incorrect	Hard	11	6	66	0.2406	0.2406	0.0000	PASS
All incorrect	Hard	1	5	5	0.2397	0.2397	0.0000	PASS
Alternating	Medium	11	6	66	0.9854	0.9854	0.0000	PASS
Mixed	Easy	4	6	24	0.9988	0.9988	0.0000	PASS
Mixed	Medium	4	6	24	0.9938	0.9938	0.0000	PASS
Mixed	Hard	3	6	18	0.8900	0.8900	0.0000	PASS
Total		**46**		**274**			**0.0000**	**PASS**

Note. Concepts in the same row received identical designed sequences, so they share the same expected final estimate. Per-concept detail is in the companion workbook.

All 46 designed cases passed. Consecutive correct sequences converged toward 1.0000, consecutive incorrect sequences settled at 0.2406, and alternating and mixed sequences produced intermediate values determined by the guess probability at each step. All 274 updates and all 46 stored final estimates fell within the ±0.01 tolerance, and every comparison agreed to less than 1×10^{-9}. In each case the prior recorded by the application equaled the independently chained prior, and the guess probability applied equaled the format-specific or rating-specific value in Table 13.

Part B: Randomly Sampled Sequences from the Implementation Period

Part B used records generated by ordinary student use. From the 46 students who generated BKT records, two per grade level were drawn at random using a published seed, and five practiced concepts were drawn for each, with

preference for concepts having two or more opportunities. Each sequence was recomputed from 0.10 using the recorded observations, following the same procedure as Part A.

Table 18.

Results of the Forty Sampled Student Sequences

Student	Sequences	Updates (Test Cases)	Sequence Length	Passed	Largest Difference	Result
G7-02	5	12	2–3	12	0.0000	PASS
G7-04	5	23	4–5	23	0.0000	PASS
G8-04	5	16	2–5	16	0.0000	PASS
G8-07	5	19	3–4	19	0.0000	PASS
G9-01	5	27	4–7	27	0.0000	PASS
G9-02	5	26	4–6	26	0.0000	PASS
G10-02	5	11	2–3	11	0.0000	PASS
G10-06	5	24	4–6	24	0.0000	PASS
Total	**40**	**158**	**2–7**	**158**	**0.0000**	**PASS**

Note. Students are identified by pseudonymized codes. Sequence Length is the range of recorded practice opportunities across the five sequences per student.

All 40 sampled sequences and all 158 updates passed within tolerance, again agreeing to less than 1×10^{-9}. The sample included sequences combining test responses and flashcard ratings within one concept, sequences beginning with a flashcard rating rather than a test response, and sequences in which the estimate decreased and later recovered, indicating that the engine applies the same equations under ordinary use as under the designed conditions of Part A.

Part C: Complete Population of Recorded Updates

Part C removed the possibility that Part B's agreement was an artifact of sampling. All 12,067 updates recorded for the 46 participating students, across 3,494 student–concept sequences, were recomputed. Two checks were applied to every update. First, the posterior was recomputed from the stored prior, the recorded observation, and the applicable parameters, then compared with the stored posterior. Second, the stored prior was verified against the stored posterior of the preceding update for that student and concept, or against 0.10 at the first update. All 12,067 updates passed both checks and applied the correct guess probability for the recorded format or rating.

Implementation Correctness Rate

Table 19 presents the Implementation Correctness Rate computed for each verification part using EQ (6), together with the agreement between the current knowledge estimate stored for each sequence and the last expected posterior computed for that sequence.

Table 19.

Correctness Rate by Verification Part

Verification Part	Sequences	Test Cases	Passed	Largest Difference	ICR (%)	Final Estimates in Agreement
A. Designed test cases executed in the live application	46	274	274	0.0000	100.00	46 of 46
B. Random sample of recorded sequences	40	158	158	0.0000	100.00	40 of 40
C. Complete population of recorded updates	3,494	12,067	12,067	0.0000	100.00	3,494 of 3,494
Parts A and B combined	**86**	**432**	**432**	**0.0000**	**100.00**	**86 of 86**

Note. One test case is one knowledge-estimate update. A case passed when the system and manual values differed by no more than ±0.01, following EQ (6).

The Implementation Correctness Rate was 100.00 percent in every part. The 432 test cases of Parts A and B, corresponding to the designed and sampled cases required by the procedure in Chapter II, all passed, and extending the comparison to the full population of 12,067 updates produced the same result. The largest absolute deviation recorded anywhere was 0.0000, indicating agreement at the limit of double-precision arithmetic rather than merely within tolerance. The stored estimate for every sequence equaled the last expected posterior with a matching opportunity count, which establishes that the values displayed to students and used to calibrate item generation and game difficulty are the values produced by the verified update sequence.

Discussion

The verification establishes that the BKT engine executes the standard four-parameter equations faithfully. Consistent with Chapter II, this concerns the correctness of the implementation, not the predictive validity of the model. A perfect correctness rate with negligible deviation is the expected outcome of this type of verification rather than an anomalous one, since the engine and the workbook evaluate the same closed-form equations in double-precision arithmetic and any material difference would indicate a defect.

The value of the procedure lies in the four specific defects it was built to catch: a guess probability that does not match the question format or flashcard rating; a missing transition phase after the evidence phase; an incorrect prior read at the start of an update; and a displayed estimate that diverges from the updates behind it. Each would have produced a deviation far exceeding the ±0.01 tolerance, and none was observed.

The results further confirm that the guess parameter configuration in Table 13 operates as specified, since the probability applied corresponds to the recorded format or rating in all 12,067 updates. This is consistent with the finding that permitting guess and slip to vary between responses produces more accurate learner models than fixed per-concept values (Šarić-Grgić et al., 2024), and confirms that the engine reflects the four-parameter formulation described in the literature (Bulut et al., 2023).

Three conditions govern interpretation. The observation entering the engine is determined by the application's grading component rather than by the researchers, and no disagreement between designed response and recorded observation occurred in Part A. The test account used in Part A is not part of the study cohort and is excluded from the records reported under Statement of the Problem No. 4. The procedure is reproducible, since Part B's sample is fixed by its published seed, Part C is deterministic, and every formula in the companion workbook can be inspected individually. Having established that the recorded estimates are computed correctly, the values stored during implementation may be treated as a dependable record of what the engine measured.

CHANGE IN THE RECORDED KNOWLEDGE ESTIMATES

This section presents the change observed in the BKT knowledge estimates recorded by LearnSync AI during the one-week usage period conducted from 7 to 13 September 2026, which constituted the second week of the three-week implementation described in Chapter II. The figures reported are the platform's own recorded output, extracted through the educators' mastery evidence export and verified for arithmetic correctness under Statement of the Problem No. 3.

What the Knowledge Estimate Measures

The engine maintains one estimate for each student and concept, representing the probability that the student has learned that concept. It changes only when a practice opportunity is recorded, which occurs when a student answers an adaptive test question or rates a due flashcard. Two values are read from each student–concept pair: the estimate after the first opportunity, taken from the earliest recorded update, and the latest estimate currently stored. A student's figure at either point is the mean across the concepts they actually practiced, and the change is the paired difference between these two means over the same concepts.

The estimate after the first opportunity is not a baseline and is not a pre-test measurement. The outcome of the first response is already contained within it, so a first correct multiple-choice answer produces 41.9 percent while a first incorrect answer produces 21.7 percent. The initialization constant of 0.10 is identical for every student, carries no information about any individual, and is never reported as a measured value. Concepts a student was never asked about are excluded from both means rather than entered at the constant, so the denominator throughout is what each student practiced. This quantity is therefore distinct from the material-level coverage measure in EQ (4), which enters unpracticed concepts at the constant and is not used here.

Participation and Volume of Practice

Table 20 presents the participation and the volume of practice recorded for each grade level during the implementation period.

Table 20.

Participation and Volume of Practice by Grade Level

Grade	Concepts	Enrolled	Started	Non-Users	Adaptive Tests	Test Questions	Card Ratings	Opportunities
Grade 7	102	8	8	0	18	1,836	799	2,635
Grade 8	97	15	14	1	33	3,201	1,449	4,650
Grade 9	46	16	15	1	34	1,564	1,042	2,606
Grade 10	70	10	9	1	21	1,470	706	2,176
All	**315**	**49**	**46**	**3**	**106**	**8,071**	**3,996**	**12,067**

Note. A practice opportunity is one recorded knowledge-estimate update. Non-users are students who never opened the material and are excluded from every mean in this section.

Forty-six of the 49 enrolled students used the platform, generating 12,067 practice opportunities across 3,494 student-concept pairs. Of these, 8,071 came from adaptive-test questions across 106 test attempts and 3,996 from flashcard ratings. Every student who started the material was assessed on all of its concepts, so the number of pairs per grade is the product of started students and concepts. Three enrolled students never used the platform and carry no estimate.

Change Across Successive Practice Opportunities

Table 21 presents the mean knowledge estimate after each successive practice opportunity on a concept, computed over every student-concept pair that reached that opportunity, together with the distribution of those pairs across the platform's three mastery tiers.

Table 21.

Knowledge Estimate After Each Practice Opportunity

Opportunity Index	Pairs Observed	Mean Estimate	Increase From Previous	Foundational	Intermediate	Advanced
1	3,494	34.4%		88.4%	11.6%	0.0%
2	3,064	58.1%	+23.7 pts	39.9%	32.2%	27.9%
3	2,508	74.7%	+16.6 pts	22.7%	22.2%	55.1%
4	1,760	83.9%	+9.2 pts	12.9%	15.8%	71.3%
5	888	88.5%	+4.6 pts	9.6%	9.2%	81.2%
6	289	89.5%	+1.0 pts	9.3%	8.0%	82.7%

Note. The opportunity index counts practice opportunities on a concept, so index 1 is the first graded one. Indices beyond six are omitted as fewer than thirty pairs reach them.

The mean estimate rose at every successive opportunity, from 34.4 percent to 89.5 percent. The increase was steepest between the first and second opportunity at 23.7 percentage points and diminished as estimates approached the upper limit of the scale, reaching 1.0 point between the fifth and sixth. The

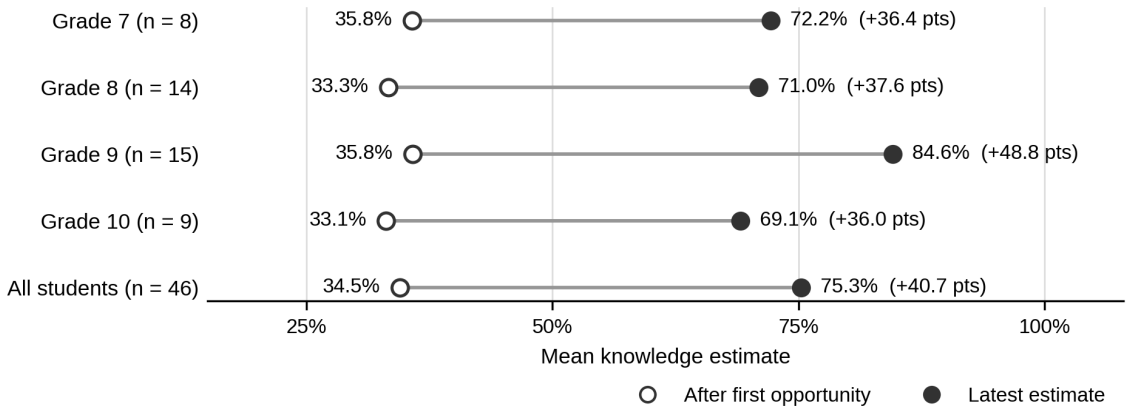
tier distribution shifted correspondingly: no pair could reach the advanced tier after a single opportunity, since the largest posterior obtainable in one step from 0.10 is 72.3 percent, whereas 71.3 percent of pairs were advanced by the fourth. These means are computed over a diminishing subset, since only 50.4 percent of pairs reached a fourth opportunity and 8.3 percent reached a sixth; the values at higher indices therefore describe the concepts students continued to practice rather than the full set.

Change per Student Over the Implementation Period

Figure 18 presents the mean knowledge estimate per student, averaged within each grade level, after the first practice opportunity and at the end of the implementation period

Figure 18.

Knowledge Estimates at the Start and End, by Grade Level



Note. n = 46 started students. Each value is the mean, within a grade level, of the students' own mean estimates over the concepts they practiced; the final row covers all started students.

Non-users are excluded. The grade levels are sampling strata; no test was conducted between them.

Table 22 presents the paired change per student, tested using a paired-samples t-test on the forty-six (46) started students and on each grade level separately. Statistical significance was evaluated at the .05 level.

Table 22.

Change in Knowledge Estimate per Student, by Grade Level

Group	n	After First Opportunity	Latest Estimate	Mean Change (pts)	Standard Deviation	Median Change	Range	t	p	dz
Grade 7	8	35.8%	72.2%	+36.4	23.1	+38.6	+1.6 to +68.6	4.45	.003	1.57
Grade 8	14	33.3%	71.0%	+37.6	17.6	+45.9	+0.3 to +58.8	8.00	< .001	2.14
Grade 9	15	35.8%	84.6%	+48.8	14.8	+53.3	+7.5 to +61.2	12.80	< .001	3.30
Grade 10	9	33.1%	69.1%	+36.0	21.4	+38.0	+0.4 to +58.5	5.04	.001	1.68
All started	**46**	**34.5%**	**75.3%**	**+40.7**	**18.8**	**+46.5**	**+0.3 to +68.6**	**14.68**	**< .001**	**2.16**

Note. Change is the latest estimate minus the estimate after the first opportunity, in percentage points, paired within students. The effect size is Cohen's d for paired samples. Each test is a within-student paired comparison; no test was conducted between grade levels.

The mean estimate across the 46 started students rose from 34.5 percent to 75.3 percent, a mean paired increase of 40.7 percentage points that was statistically significant, $t(45) = 14.68$, $p < .001$, with a very large effect size of $dz =$

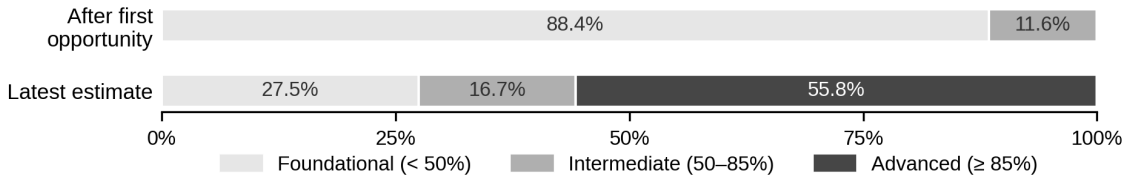
2.16. The increase was significant within every grade level. Every started student's mean rose, with individual changes ranging from 0.3 to 68.6 percentage points. The mean change ranged from 36.0 to 48.8 percentage points across the four grade levels. Because the study did not test for differences between grade levels, these values are reported as stratum-level descriptions rather than as evidence that any grade level responded differently to the platform.

Movement Between Knowledge Tiers

Figure 19 presents the distribution of the three thousand four hundred ninety-four (3,494) student–concept pairs across the platform's knowledge tiers at the two measurement points.

Figure 19.

Movement Between Knowledge Tiers



Note. n = 3,494 student–concept pairs. Tier thresholds are those applied by the platform, namely 50 percent for the intermediate tier and 85 percent for the advanced tier.

The share of pairs at the advanced tier rose from 0.0 percent to 55.8 percent, while the foundational share fell from 88.4 percent to 27.5 percent. The intermediate share rose modestly from 11.6 to 16.7 percent, reflecting its position as a band pairs pass through rather than accumulate in. The advanced share

necessarily begins at zero because a single opportunity cannot carry an estimate above the 85 percent threshold from the initialization constant.

Practice Volume and the Magnitude of Change

Table 23 groups the started students by the size of their change and reports the volume of practice recorded in each group.

Table 23.

Practice Volume by Size of Change

Magnitude of Change	Students	Mean Change (pts)	Mean Adaptive Tests Taken	Mean Card Ratings	Mean Practice Opportunities
25.0 points or more	36	+49.0	2.6	103	294
10.0 to 24.9 points	5	+18.6	1.6	42	200
5.0 to 9.9 points	2	+7.1	1.0	18	76
Less than 5.0 points	3	+0.8	1.0	15	105

Note. n = 46 started students. Values are means within each group. Students in the lowest group practiced most concepts only once, so no change can be observed for those concepts.

The magnitude of change tracked the number of adaptive tests and card ratings recorded. Thirty-six students gained at least 25.0 percentage points, averaging 2.6 adaptive tests and 103 flashcard ratings. The five who gained between 10.0 and 24.9 points and the two who gained between 5.0 and 9.9 had taken fewer tests and recorded fewer ratings. The three whose change was

below 5.0 points had each taken a single adaptive test and recorded fewer than 20 ratings, leaving most of their concepts with only one opportunity. Taken together, five of the 46 started students recorded changes below 10.0 percentage points, and three enrolled students generated no records at all, so a comparable change was not observed for every student enrolled.

Discussion

The recorded estimates increased consistently across the period, both within concepts as opportunities accumulated and within students as overall practice accumulated. The pattern in Table 21, where the increase is largest at the second opportunity and diminishes toward the upper end of the scale, is the behavior expected of the BKT update: each observation is weighted against the current estimate, so the same evidence produces a smaller adjustment once the estimate is already high. The uniform direction of change across all 46 started students, with a very large effect size, indicates that the recorded practice was sufficient to move the model's estimates for every student who used the platform.

Four conditions govern interpretation. First, the estimate after the first opportunity is not a pre-test score and the design is not a pre-test and post-test comparison; a one-week usage period is too short to attribute a difference in measured performance to the platform, so this section reports the trajectory the model recorded rather than a claim of learning gain. Second, the estimate is model output under the parameters stated in Chapter II rather than a test score,

and its arithmetic correctness, though not its predictive validity, was established under Statement of the Problem No. 3. Third, the model contains no forgetting term, so an estimate can move only when an opportunity is recorded, meaning an unchanged estimate indicates absence of practice rather than absence of learning. Fourth, each student is measured on the concepts they practiced, so these figures are not comparable with the coverage measure in EQ (4).

The relationship between practice volume and magnitude of change in Table 23 is consistent with the structure of the model, in which the transition parameter is applied only when an opportunity is recorded. Students in the largest-change group recorded the most adaptive tests and card ratings, while those whose concepts were practiced only once showed little or no change. Because the design includes no comparison group and does not control for differences among the students themselves, this relationship is reported as an observed association rather than as evidence that practice volume alone produced the change. The students' and educators' assessment of the platform following this implementation period is reported under Statement of the Problem No. 5.

LEVEL OF ACCEPTABILITY OF THE PLATFORM

This section reports how the users who evaluated the platform assessed its quality at the end of the implementation period, in terms of the eight ISO/IEC 25010 characteristics identified in Chapter I: functional suitability, performance

efficiency, compatibility, usability, reliability, security, maintainability, and portability. The same model has been applied in a recent Philippine evaluation of educational software rated by a panel of ten expert evaluators (Alan et al., 2026).

Respondents and Instrument

The questionnaire was administered through Google Forms after the implementation period. Fifty-seven (57) respondents completed it: forty-six (46) Junior High School students, one (1) educator, and ten (10) information technology practitioners who evaluated the platform as technical experts. The students and the educator rated only the functionalities accessible through their own portals, and their number corresponds to the forty-six (46) students and one (1) educator whose recorded use is reported under Statement of the Problem No. 4. The instrument contained nineteen (19) close-ended statements distributed across the eight characteristics, plus one open-ended item inviting suggestions. The statements used the four-point Likert scale in Table 2, which removes the neutral option to encourage a definitive evaluation. All fifty-seven (57) respondents answered every close-ended statement, so no responses were excluded.

Responses were analyzed using frequency count, weighted mean, standard deviation, and the verbal interpretation ranges in Table 2, with each mean rounded to two decimal places before its range was determined. Cronbach's alpha computed over the nineteen (19) statements yielded 0.397 for

the fifty-seven (57) responses. This falls below the .70 convention applied to the requirements analysis instrument and is addressed in the discussion at the end of this section. Because the three respondent groups evaluated the platform from different vantage points, the results are reported both in aggregate and by group.

Level of Acceptability by Quality Characteristic

Table 24 presents the frequency distribution, weighted mean, standard deviation, and verbal interpretation of each of the nineteen (19) statements, grouped by the quality characteristic they measure, together with the composite mean of each characteristic.

Table 24.

Acceptability Ratings by Indicator and Quality Characteristic

Indicator	**Frequency of Responses**				**Weighted Mean**	**Standard Deviation**	**Verbal Interpretation**
	4	**3**	**2**	**1**			
1. The app performs the tasks it is supposed to do.	28	21	5	3	3.30	0.84	Strongly Agree
2. The app gives correct and accurate results.	27	23	4	3	3.30	0.82	Strongly Agree
Composite Mean - Functional Suitability					**3.30**	**0.63**	**Strongly Agree**
3. The app responds quickly when I tap or click something.	1	5	24	27	1.65	0.72	Strongly Disagree

4. The app uses resources efficiently.	2	13	28	14	2.05	0.79	Disagree
Composite Mean - Performance Efficiency					**1.85**	**0.53**	**Disagree**
5. The app works well on my device (phone, tablet, or computer).	25	22	4	6	3.16	0.96	Agree
6. The app works smoothly without conflicting with other apps I use.	25	22	8	2	3.23	0.82	Agree
Composite Mean - Compatibility					**3.19**	**0.69**	**Agree**
7. It was easy to understand how to use the app.	28	21	6	2	3.32	0.81	Strongly Agree
8. It was easy to learn how to use the app.	24	21	9	3	3.16	0.88	Agree
9. The app was easy and comfortable to use overall.	18	25	10	4	3.00	0.89	Agree
10. The app looks good and pleasant to use.	24	23	10	0	3.25	0.74	Agree
Composite Mean - Usability					**3.18**	**0.41**	**Agree**
11. The app works without crashing or freezing unexpectedly.	26	21	4	6	3.18	0.97	Agree
12. If something goes wrong, the app recovers well and keeps my data safe.	22	23	5	7	3.05	0.99	Agree
Composite Mean - Reliability					**3.11**	**0.74**	**Agree**
13. The app keeps my personal information safe.	28	19	8	2	3.28	0.84	Strongly Agree

14. The app prevents other people from seeing or changing my data.	22	28	5	2	3.23	0.76	Agree
15. I feel my account is safe from hackers or unauthorized users.	32	18	5	2	3.40	0.80	Strongly Agree
Composite Mean - Security					**3.30**	**0.47**	**Strongly Agree**
16. If there is a problem, it seems easy to find where it went wrong.	29	20	4	4	3.30	0.89	Strongly Agree
17. The app seems easy to test and check for errors.	22	26	7	2	3.19	0.79	Agree
Composite Mean - Maintainability					**3.25**	**0.61**	**Agree**
18. The app was easy to install or access.	24	26	5	2	3.26	0.77	Strongly Agree
19. The app could easily replace other similar apps or tools I have used.	24	23	7	3	3.19	0.85	Agree
Composite Mean - Portability					**3.23**	**0.58**	**Agree**
Overall Composite Mean					**3.08**	**0.24**	**Agree**

Note. n = 57. The table shows how the 57 evaluators answered the nineteen statements, grouped by quality characteristics. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

Seventeen (17) of the nineteen (19) statements obtained a verbal interpretation of Agree or Strongly Agree, of which seven (7) were interpreted as Strongly Agree. The highest-rated statement was the perceived safety of the account from unauthorized users at 3.40, followed by the ease of understanding how to use the platform at 3.32. The two statements under performance efficiency were the only ones rated below the agreement threshold. The speed of

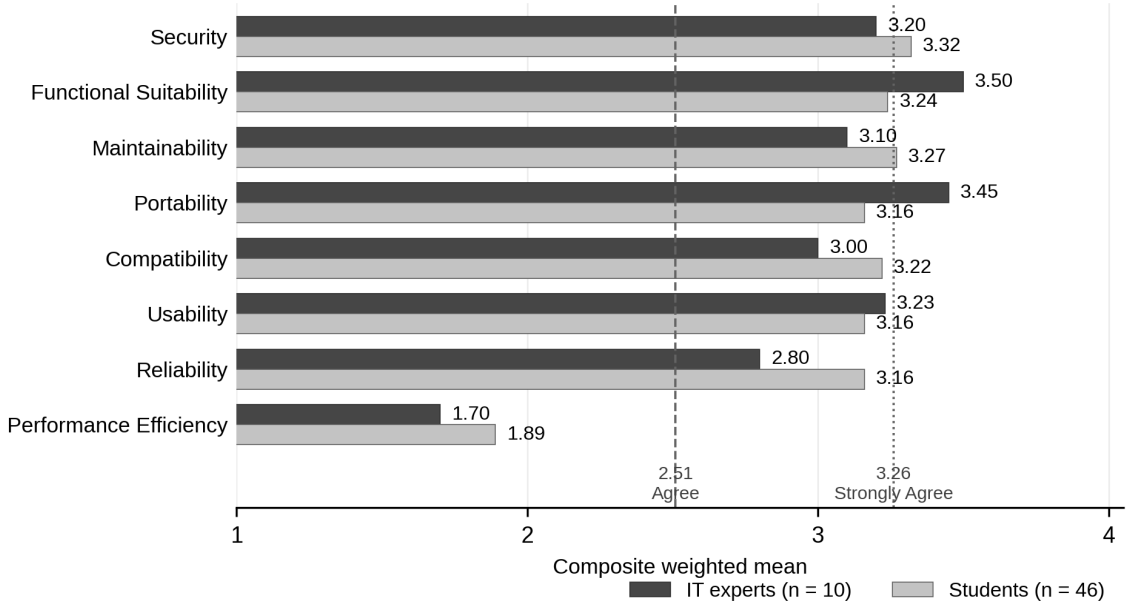
the platform's response to a tap or click obtained the lowest weighted mean of any statement at 1.65, interpreted as Strongly Disagree, with fifty-one (51) of the fifty-seven (57) respondents disagreeing or strongly disagreeing, while the efficient use of device resources obtained 2.05, interpreted as Disagree, with forty-two (42) respondents disagreeing or strongly disagreeing.

Results by Respondent Group

Figure 20 compares the composite mean of each quality characteristic between the students and the information technology practitioners, and Table 25 presents the composite means of all three respondent groups together with the aggregate result and its rank.

Figure 20.

Quality Ratings by Respondent Group



Note. n = 57. The educator is not shown because that group has only one respondent; those values are in Table 25. The dashed line marks 2.51, the start of Agree, and the dotted line marks 3.26, the start of Strongly Agree.

Table 25.

Quality Ratings by Respondent Group and Rank

Quality Characteristic	Items	Students (n = 46)	Educator (n = 1)	IT Experts (n = 10)	All (n = 57)	Verbal Interpretation	Rank
Security	3	3.32	3.67	3.20	3.30	Strongly Agree	1
Functional Suitability	2	3.24	4.00	3.50	3.30	Strongly Agree	2
Maintainability	2	3.27	3.50	3.10	3.25	Agree	3
Portability	2	3.16	4.00	3.45	3.23	Agree	4
Compatibility	2	3.22	4.00	3.00	3.19	Agree	5
Usability	4	3.16	3.50	3.23	3.18	Agree	6
Reliability	2	3.16	4.00	2.80	3.11	Agree	7

Quality Characteristic	Items	Students (n = 46)	Educator (n = 1)	IT Experts (n = 10)	All (n = 57)	Verbal Interpretation	Rank
Performance Efficiency	2	1.89	1.50	1.70	1.85	Disagree	8
Overall Composite Mean	**19**	**3.08**	**3.53**	**3.03**	**3.08**	**Agree**	

Note. n = 57. The table shows the composite mean of each quality characteristic for each group of evaluators. Ranks use the unrounded means, so Security is ranked ahead of Functional Suitability although both display as 3.30.

Seven (7) of the eight (8) quality characteristics obtained a verbal interpretation of Agree or Strongly Agree in the aggregate. Security ranked highest at 3.30, followed closely by functional suitability, also at 3.30, both interpreted as Strongly Agree. Maintainability, portability, compatibility, usability, and reliability followed, with composite means between 3.11 and 3.25, all interpreted as Agree. Performance efficiency was the only characteristic rated below the agreement threshold, obtaining 1.85 and ranking eighth. The overall composite mean of the nineteen (19) statements was 3.08 with a standard deviation of 0.24, interpreted as Agree, indicating that the evaluators found LearnSync AI acceptable overall.

The three respondent groups reached broadly consistent conclusions. The students returned an overall composite mean of 3.08 and the information technology practitioners 3.03, while the single educator returned 3.53. All three groups placed performance efficiency lowest by a wide margin, rating it 1.89, 1.70, and 1.50 respectively, and all three rated every other characteristic within or above the agreement range. Where the groups differed, the differences were

characteristic-specific rather than systematic: the practitioners rated functional suitability and portability higher than the students did, at 3.50 and 3.45 against 3.24 and 3.16, while rating reliability and compatibility lower, at 2.80 and 3.00 against 3.16 and 3.22.

Suggestions for Improvement

Table 26 presents the categories into which the responses to the open-ended item were grouped.

Table 26.

Categories of Suggestions for Improvement

Category of Suggestion	Frequency	Percentage
No suggestion offered	39	68.4
Speed and stability of the platform	6	10.5
Interface clarity and ease of navigation	5	8.8
Accuracy of search results	2	3.5

Category of Suggestion	Frequency	Percentage
Availability of offline access	2	3.5
Login and data synchronization	2	3.5
Timeliness of notifications	1	1.8
Total	**57**	**100.0**

Note. n = 57. Replies such as "none" and not at the "moment" are counted as no suggestion. Percentages are of all 57 respondents. Of the 18 suggestions given, 16 came from students and 2 from the IT practitioners.

Thirty-nine (39) respondents, or 68.4 percent, offered no suggestion for improvement. Among the eighteen (18) who did, the most frequently raised concern was the speed and stability of the platform, cited by six (6) respondents, followed by the clarity of the interface and the ease of locating features, cited by five (5). The remaining suggestions concerned the accuracy of search results, the availability of offline access, the login and synchronization process, and the timeliness of notifications. That speed and stability formed the largest category of substantive suggestion, and that it was raised by both students and practitioners, corroborates the quantitative result obtained for performance efficiency.

Discussion

The evaluation indicates that the intended users found LearnSync AI acceptable, with seven (7) of the eight (8) ISO/IEC 25010 characteristics interpreted as Agree or Strongly Agree. The two highest-rated characteristics, security and functional suitability, correspond respectively to the row-level

security policies described in Chapter II, which isolate each student's data and restrict educator access to their own classes, and to the platform performing the tasks it was built to perform, which is consistent with the technical correctness established under Statement of the Problem No. 3.

Performance efficiency was the single characteristic rated below the agreement threshold, and the perceived responsiveness of the interface obtained the lowest weighted mean of any statement. This result is corroborated by the open-ended responses, in which speed and stability formed the second most frequently raised category of suggestion. It is consistent with the deployment conditions of the implementation period, during which the platform performed generative operations such as material parsing, adaptive item generation, and the semantic evaluation of Free Recall and Feynman responses through external model calls, each of which introduces latency visible to the user at the point of interaction. Performance efficiency is therefore identified as the principal area for improvement in the developed platform.

That all three respondent groups independently placed performance efficiency lowest strengthens the finding. The students and the educator used the platform over the implementation week on their own devices and network connections, while the information technology practitioners assessed it as a software artifact, and the three vantage points produced the same verdict on responsiveness. The practitioners' own suggestions, which concerned memory consumption during extended sessions and caching to reduce load times on slower devices, point to the same mechanism identified above.

Two key factors are essential for understanding these results. First, the survey questions do not measure a single, unified concept. The reliability score was low (0.397), largely because the questions on performance efficiency did not align with the others. Because of this, the individual category scores in Table 25 should be used to evaluate quality rather than one combined overall score. Second, this evaluation reflects subjective user impressions rather than actual computational accuracy (which is evaluated separately under Statement of the Problem No. 3). Additionally, participants only reviewed features they could access; because only one of the 57 participants was a teacher, the educator portal represents just a single evaluation.

CHAPTER IV

INTRODUCTION

CONCLUSIONS

RECOMMENDATIONS

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APPENDIX A : OPERATIONAL DEFINITION OF TERMS

OPERATIONAL DEFINITION OF TERMS

Adaptive Flashcards - Digital review cards within the platform that adjust reappearance intervals (1, 3, or 7 days) and update BKT knowledge probabilities based on a student's self-reported confidence rating (Easy, Medium, or Hard).

Adaptive Test Mode - An automated assessment module that dynamically scales item difficulty (foundational, intermediate, or advanced) based on a student's real-time BKT knowledge score and calibrates scoring using format-specific guess probabilities.

Agile Scrum Framework - The iterative software engineering paradigm utilized to structure project management into distinct sprint cycles, backlogs, and rapid testing increments throughout system development.

Bayesian Knowledge Tracing - A probabilistic hidden Markov model implementing four parameters: initial Knowledge, Learning Rate, Slip Probability, and Guess Probability, to continuously track a student's knowledge of specific concepts in real time.

Black-Box Functional Testing - A software verification approach used to validate system outputs against specified inputs across the Educator and Student portals without inspecting internal backend source code.

Change in Knowledge Estimate - The arithmetic difference between a student's final knowledge estimate and initial knowledge estimate for a given learning topic, computed at the end of the implementation period.

Cognitive Load Theory - An educational design framework that guides the platform's centralized layout to reduce extraneous mental effort caused by tool fatigue and multi-platform navigation.

Cosine Similarity - A mathematical calculation executed natively within the database using the Supabase pgvector extension to measure the semantic alignment between a student's typed Free Recall summary and the stored lesson embeddings.

Digital Badges - Gamified milestone rewards awarded to students on their progress dashboard upon reaching specific study achievements or knowledge thresholds.

Document Knowledge Score - The dynamic arithmetic average of all individual concept BKT knowledge probabilities within an uploaded document, used to display overall progress and calibrate game difficulty.

Educator Portal - The secure interface enabling authenticated teachers to upload PDF learning modules or textbook scans, assign grade-level viewing access, and monitor student knowledge analytics.

Feynman Technique - An AI-assisted study feature that guides an LLM to evaluate student explanations written in plain language by conducting jargon checks, verifying analogy use, and identifying conceptual gaps.

Final Knowledge Estimate - The Bayesian Knowledge Tracing knowledge probability recorded for a student in a given learning topic after the last practice opportunity completed within the implementation period.

Free Recall - A cognitive study mode where students type out lesson recollections without notes or prompts, evaluated automatically via semantic vector comparison.

Generative Artificial Intelligence - The artificial intelligence pipeline integrated into the platform to generate adaptive test items, flashcards, gamified content, and feedback derived exclusively from educator-uploaded materials.

Guess Probability - A parameter in Bayesian Knowledge Tracing representing the probability that a student answers an item correctly through random guessing, calibrated dynamically by question format and self-reported confidence level.

Implementation Correctness Rate - A software verification metric that quantifies the percentage of system BKT outputs that match hand-calculated expected values within a +/- 0.01 floating-point tolerance.

Initial Knowledge State - A Bayesian Knowledge Tracing parameter set to a baseline default of 0.10, representing the starting prior probability that a student has learned a concept before completing any activity.

ISO/IEC 25010 Software Quality Model - The international standard framework used to evaluate LearnSync AI across eight software quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Knowledge Estimate - The Bayesian Knowledge Tracing probability, denoted as P(L), that a student has learned a given concept, recomputed by the system after every practice opportunity based on the student's response.

Knowledge Threshold - The Bayesian Knowledge Tracing probability at or above which a learning topic is considered learned by a student. This study adopts a threshold of 0.85.

Learning Progress - The change and trajectory of a student's Bayesian Knowledge Tracing knowledge estimates across successive practice opportunities within the implementation period, described in terms of the

initial knowledge estimate, the final knowledge estimate, the change between them, and the attainment of the knowledge threshold.

Learning Rate - A Bayesian Knowledge Tracing parameter set to a default value of 0.20, representing the probability that a student transitions from an unlearned to a learned state following a study opportunity.

LearnSync AI - The web-based personalized learning companion system designed for Junior High School students at Pasig Christian Academy.

Leitner System - A review framework integrated into Adaptive Flashcards that categorizes review items into progressive review tiers based on recall performance.

Optical Character Recognition - The image-processing technology used in the ingestion pipeline to convert scanned printed textbook pages into machine-readable text.

pgvector - An open-source PostgreSQL extension integrated into Supabase to store high-dimensional embeddings and execute vector distance calculations natively within the database tier.

PlayMe Portal - The gamified review hub featuring Crosswords, Term Matching, and Timed Trivia, where game difficulty and timer constraints are calibrated using the student's aggregated document knowledge score.

Practice Opportunity - A single scored student response within Adaptive Flashcards or Adaptive Test Mode that triggers an update of the Bayesian Knowledge Tracing knowledge probability for the associated learning topic. Interactions with AI-generated educational games, the AI Learning Assistant, and the AI-assisted study strategies do not constitute practice opportunities.

Proportionate Stratified Random Sampling - The probability sampling technique used to select 49 Junior high School student participants proportionally across Grades 7 through 10.

Retrieval Practice - A cognitive study strategy operationalized through quizzes, flashcards, and games that requires learners to reconstruct knowledge from memory rather than passively rereading text.

Row-Level Security - Database policy rules configured in PostgreSQL to isolate individual student data and restrict educator access strictly to their assigned grade-level classes.

Self-Regulated Learning - The process of self-directed study supported by platform analytics, enabling students to set goals, monitor knowledge curves, and adjust study habits without direct teacher supervision.

Slip Probability - A Bayesian Knowledge Tracing parameter set to 0.15, representing the probability that a student who has learned a concept still responds incorrectly due to carelessness, misreading, or fatigue.

Spaced Repetition - A scheduling strategy based on the Leitner system that determines when flashcards reappear for review based on prior recall difficulty.

Student Portal - The web application tier through which Junior High School students access study tools, complete adaptive quizzes, play review games, and view their personalized progress dashboards.

Study Streak - A gamification feature tracking consecutive daily study activity to foster consistent, habit-forming independent learning behavior.

Vector Embeddings - High-dimensional numerical representations of text generated from uploaded study resources, stored in PostgreSQL via pgvector for semantic search and AI retrieval.

APPENDIX B : EVALUATION SHEETS

EVALUATION SHEETS

APPENDIX C : PROJECT TIMELINE

PROJECT TIMELINE

APPENDIX D : WORK ASSIGNMENTS

WORK ASSIGNMENTS

Baruelo, John Louis E. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
1	The Problem and its Background	Crossword	Crossword puzzle based on the concept
1	Significance of the Study	Time Trivia	Quiz based game but with time pressure
1	Statement of the Problem	Matching Card	Matching terms and connect them with each other
2	Software Engineering Paradigm	Admin	Grade Year Level Assignments, Account Creation
2	Population and Sampling		

Leonidas, Ella Mae C. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
1	Review of Related Literature	Student Dashboard	Tracking progress, knowledge level, other info
1	Scope and Limitations	Educator Dashboard	Overall tracking progress, knowledge level of each students, other info
2	Evaluation Instrument		
2	Data Collection Method		
2	Ethical Considerations		

Soria, Marc Angelo A. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
2	Introduction	Flashcards	Spaced Repetition System, creation of flashcards
2	Research Design	Feynman	Understand the concept in a simpler one, AI to generate a simpler concept
2	Architectural Framework	Free Recall	Write all of the concepts

			memorized, AI will check the written concepts
2	Algorithm	Test Mode	Overall understanding of concepts to test
2	Tool to Test for Accuracy	BKT knowledge	Knowledge calculations
2	Summary	Educator Upload	Document extraction

APPENDIX E : CERTIFICATE OF APPRECIATION

CERTIFICATE OF APPRECIATION

APPENDIX F : SOURCE CODE

SOURCE CODE

APPENDIX G : PLAGIARISM CHECK RESULTS

PLAGIARISM CHECK RESULTS

APPENDIX H : AI GENERATED CONTENT CHECK RESULTS

AI GENERATED CONTENT CHECK RESULTS

APPENDIX I : EXTENDED ABSTRACT

EXTENDED ABSTRACT

CURRICULUM VITAE